

BFS Traversal

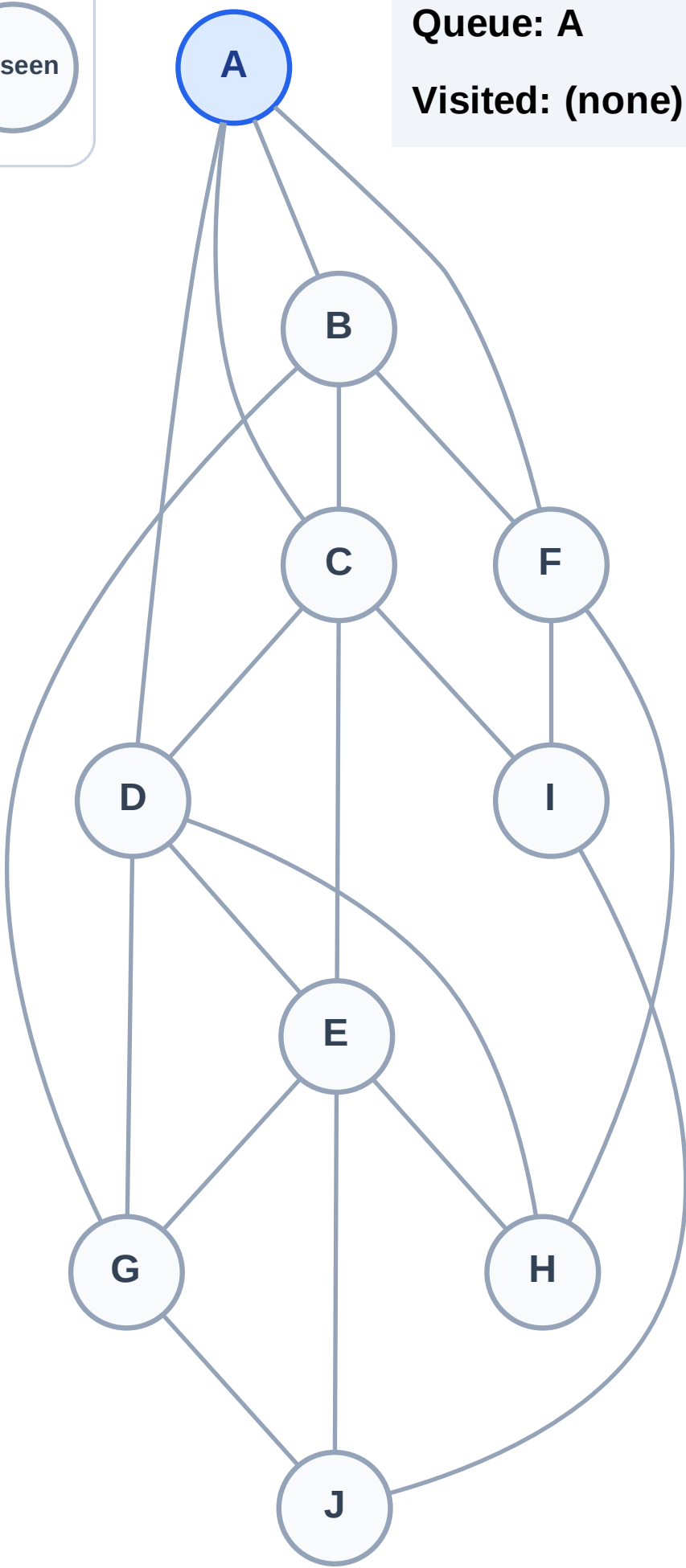
Step 1: Start at A

Legend



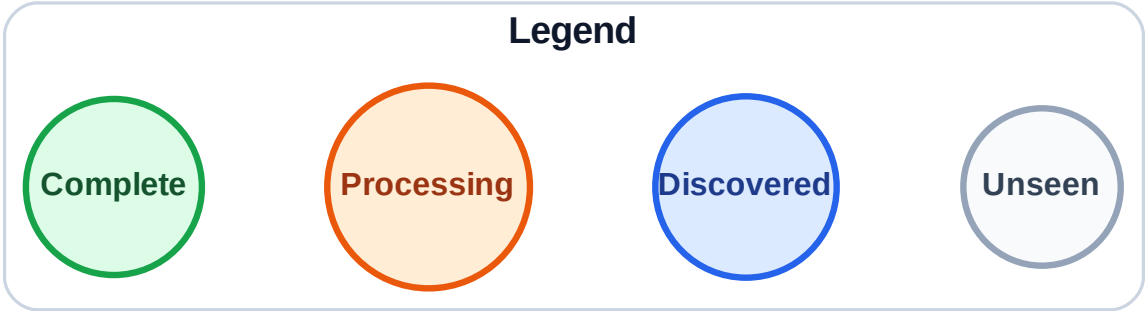
Queue: A

Visited: (none)

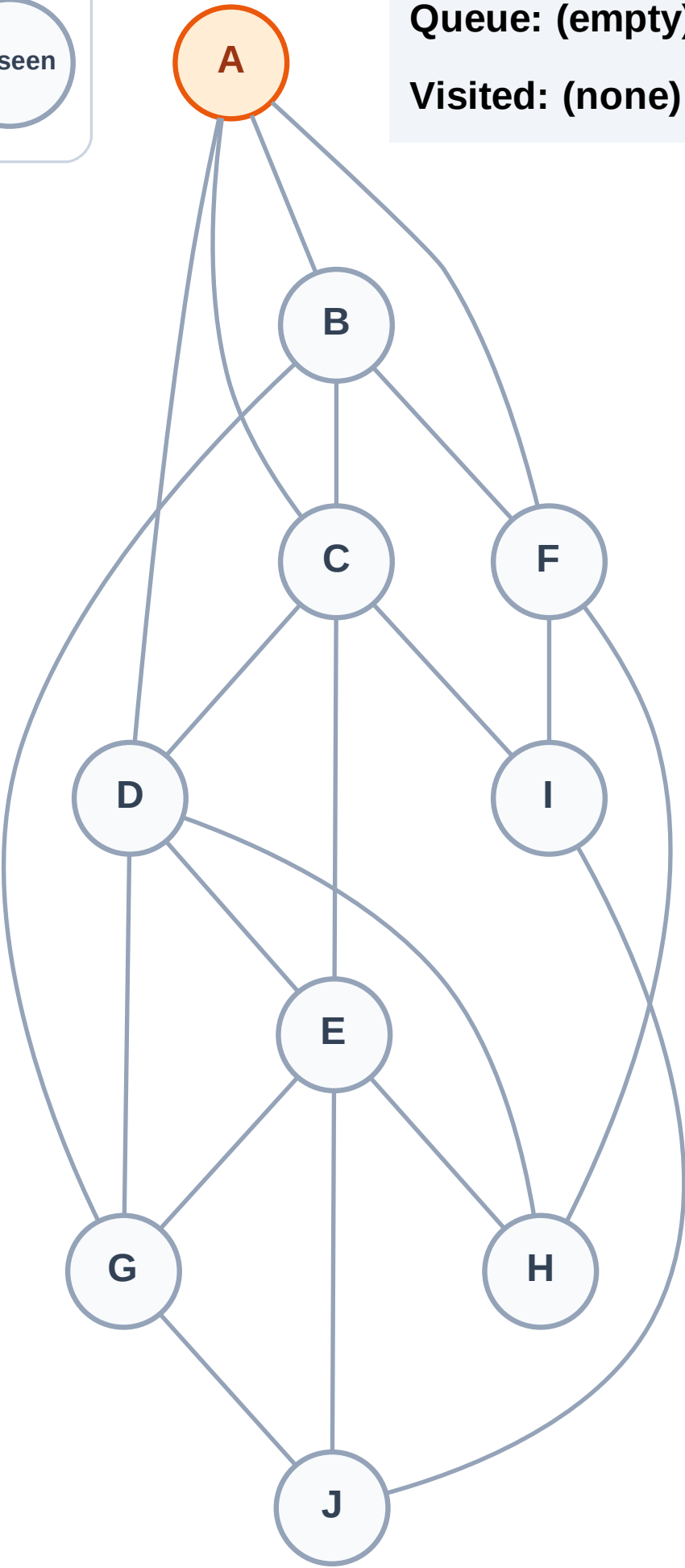


BFS Traversal

Step 2: Process node A



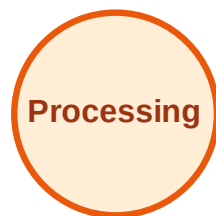
Queue: (empty)
Visited: (none)



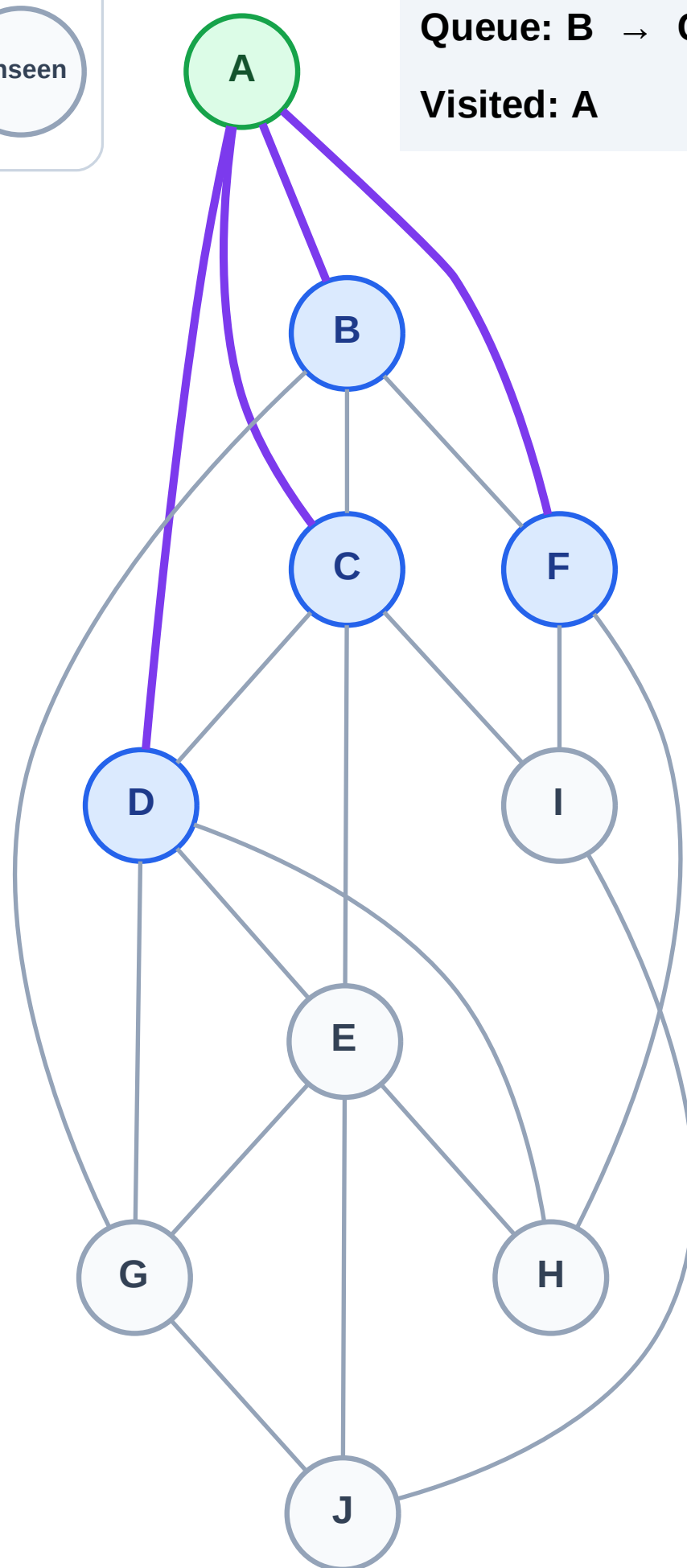
BFS Traversal

Step 3: Discover B, C, D, F from A

Legend



Queue: B → C → D → F
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

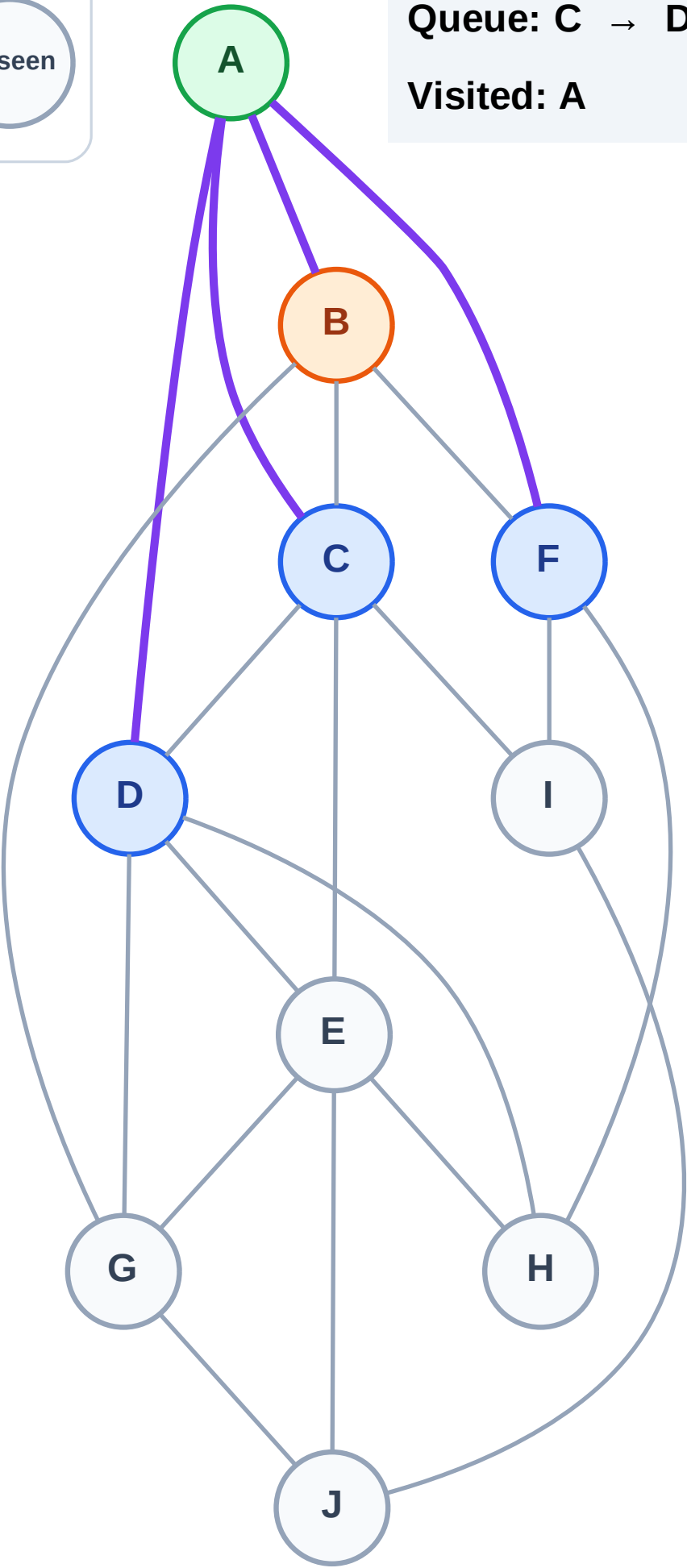
Processing

Discovered

Unseen

Queue: C → D → F

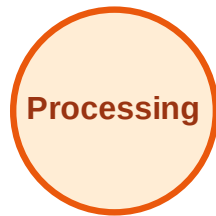
Visited: A



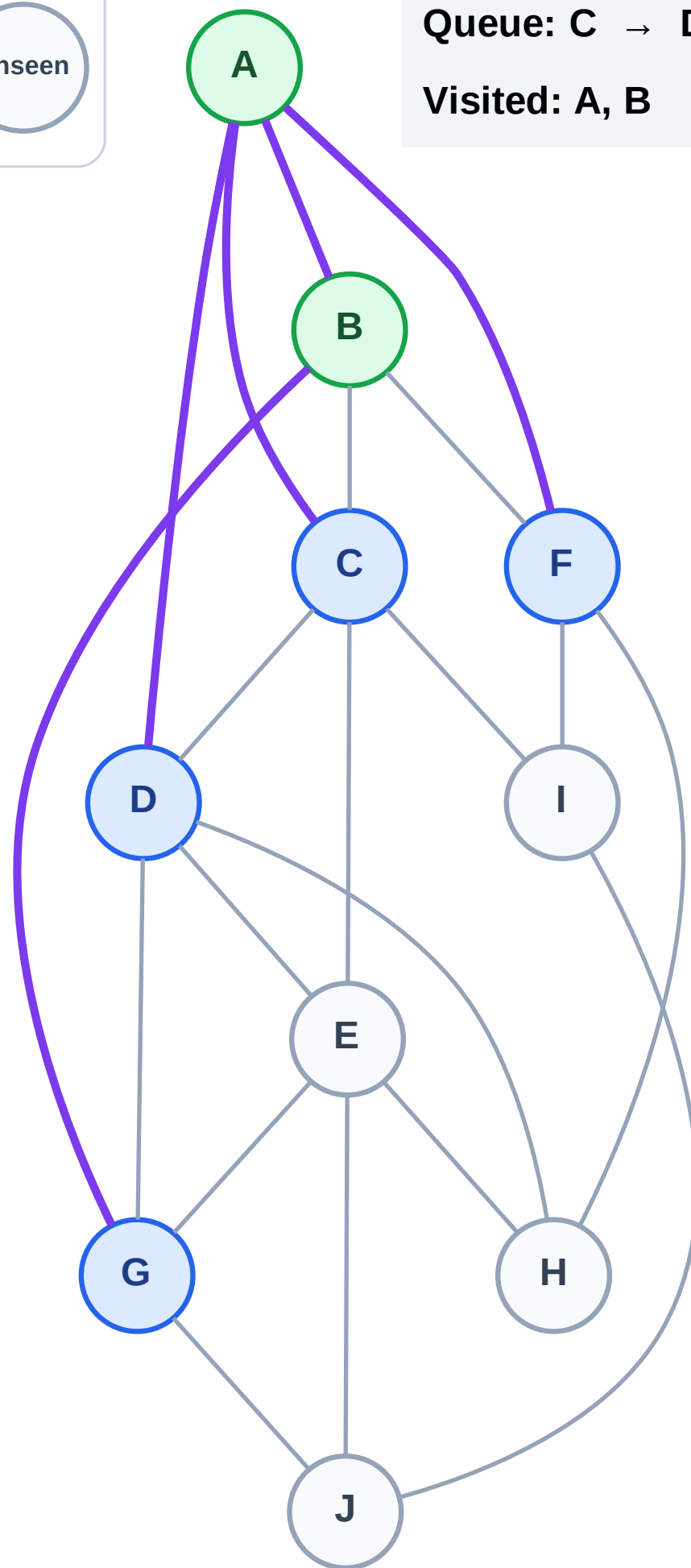
BFS Traversal

Step 5: Discover G from B

Legend



Queue: C → D → F → G
Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

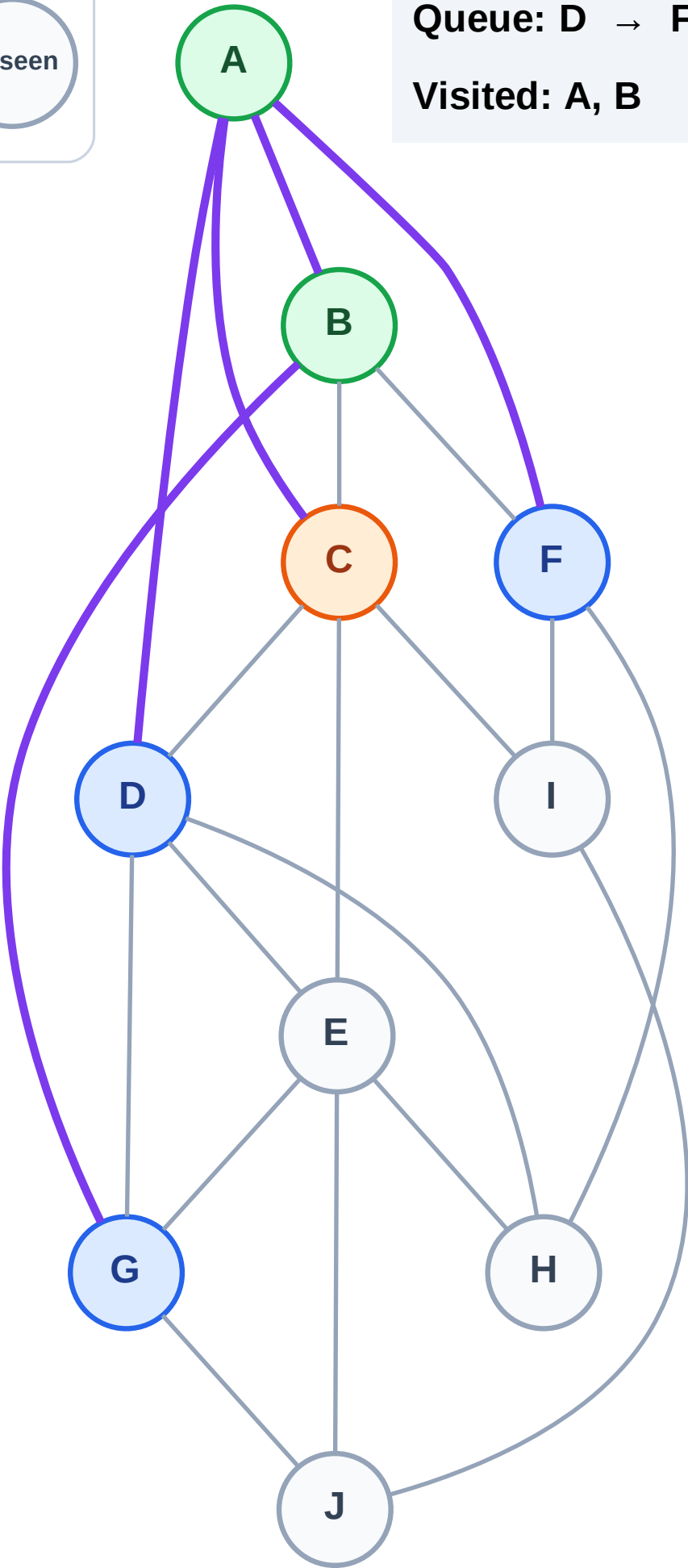
Processing

Discovered

Unseen

Queue: D → F → G

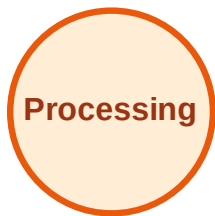
Visited: A, B



BFS Traversal

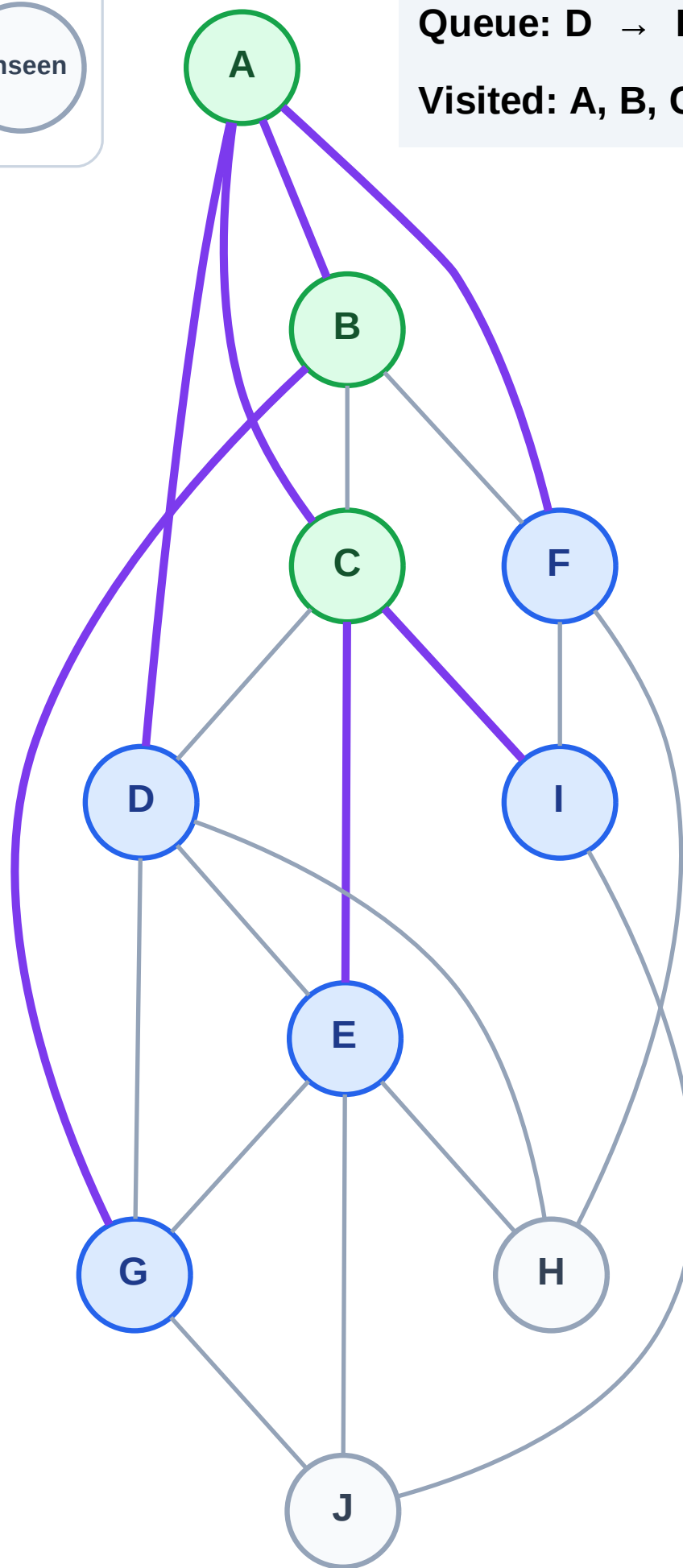
Step 7: Discover E, I from C

Legend



Queue: D → F → G → E → I

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

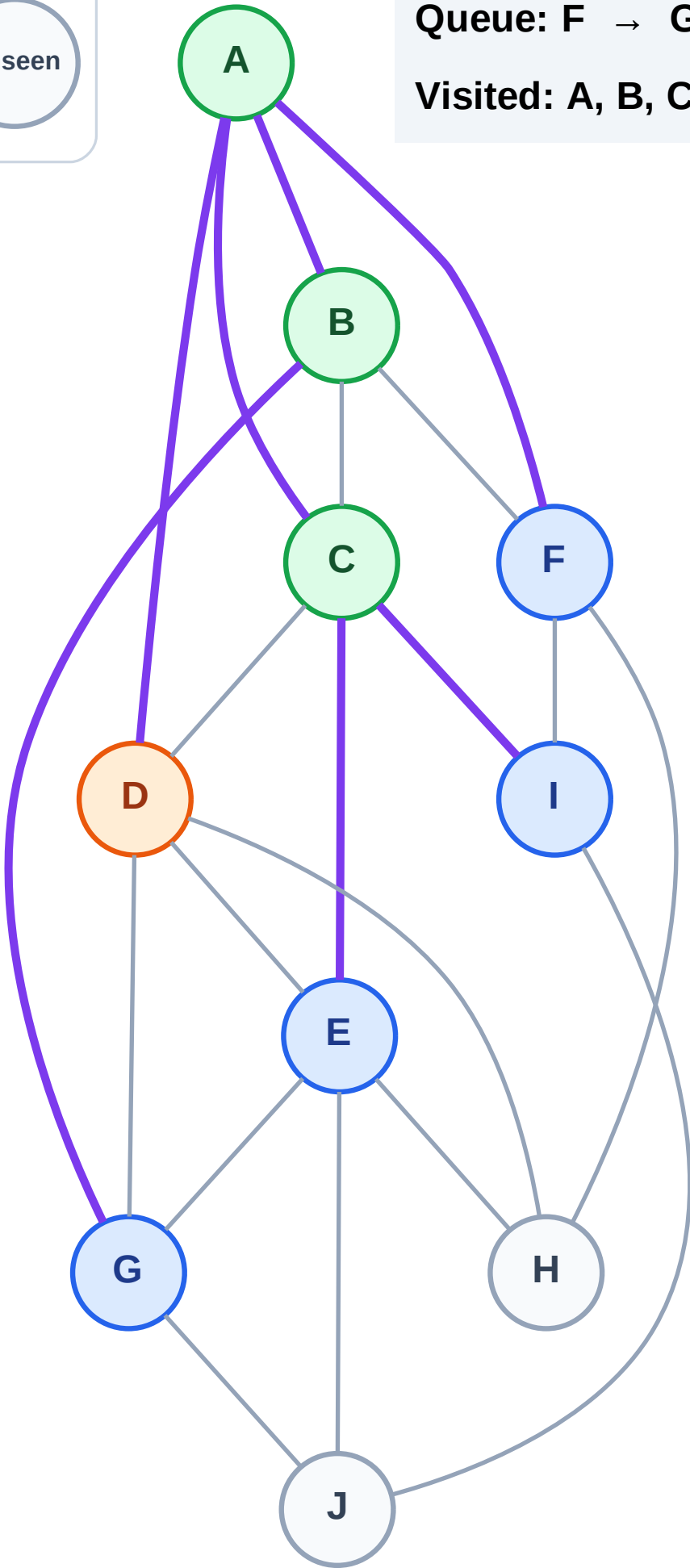
Processing

Discovered

Unseen

Queue: F → G → E → I

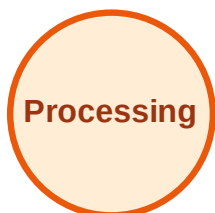
Visited: A, B, C



BFS Traversal

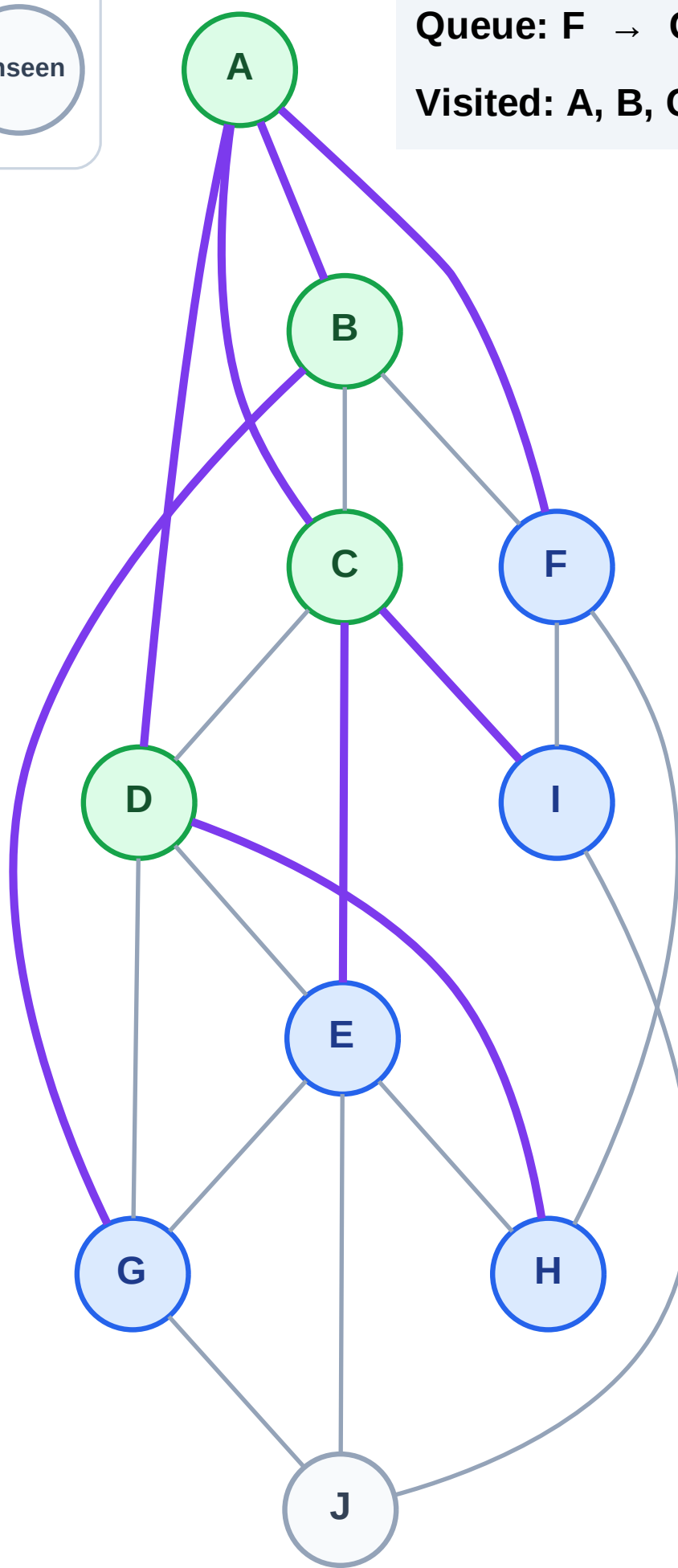
Step 9: Discover H from D

Legend



Queue: F → G → E → I → H

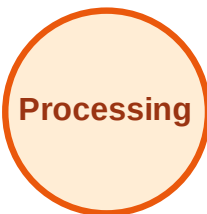
Visited: A, B, C, D



BFS Traversal

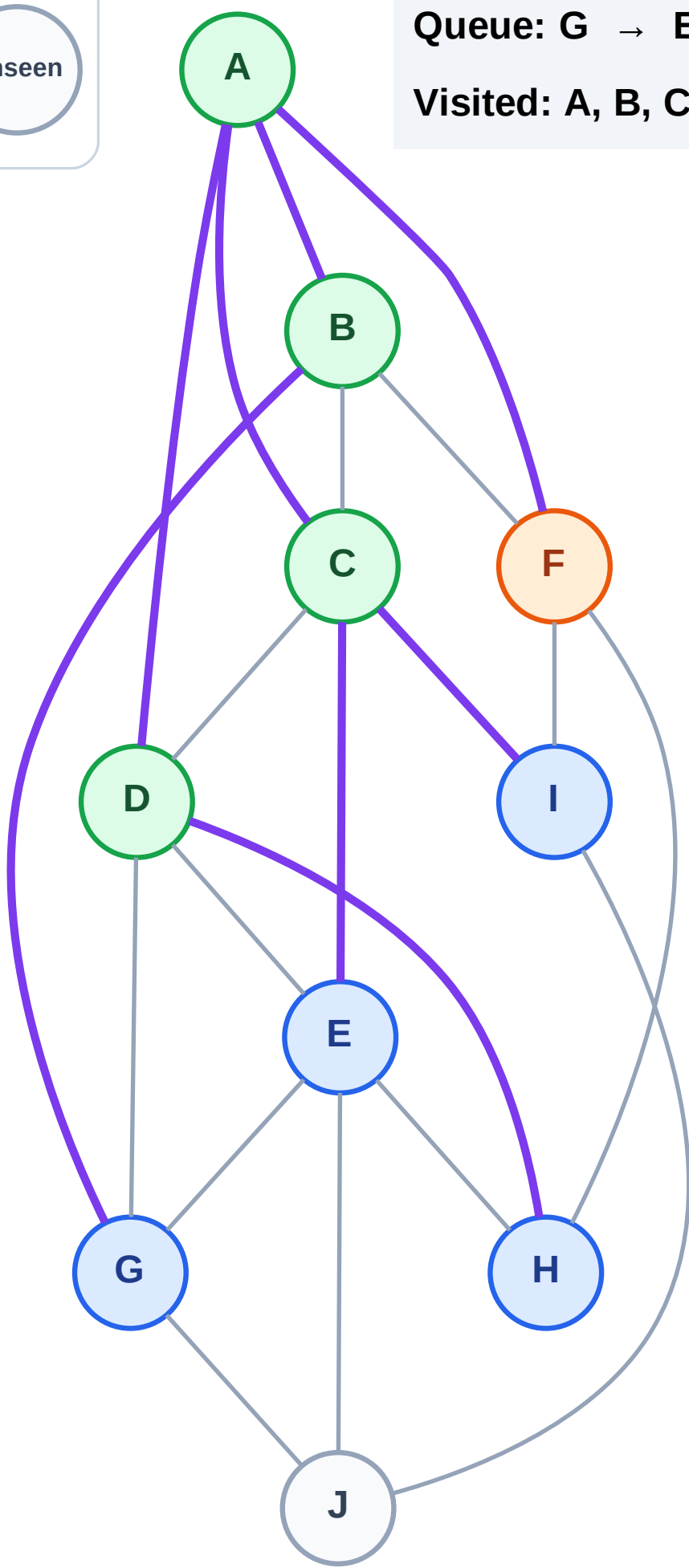
Step 10: Process node F

Legend



Queue: G → E → I → H

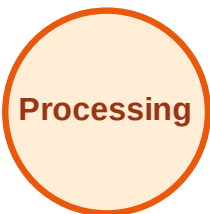
Visited: A, B, C, D



BFS Traversal

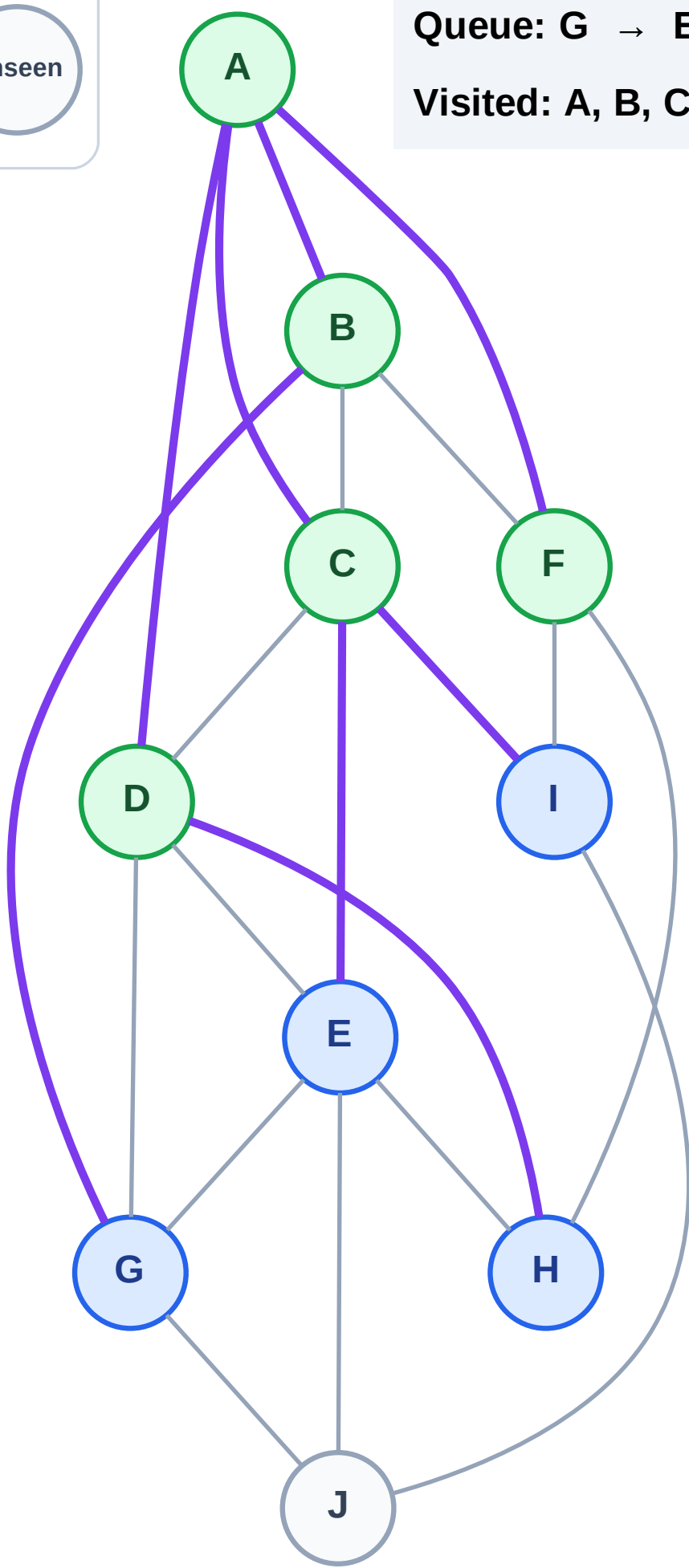
Step 11: No new neighbors from F

Legend



Queue: G → E → I → H

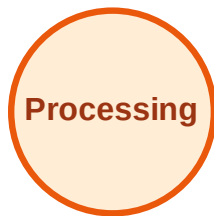
Visited: A, B, C, D, F



BFS Traversal

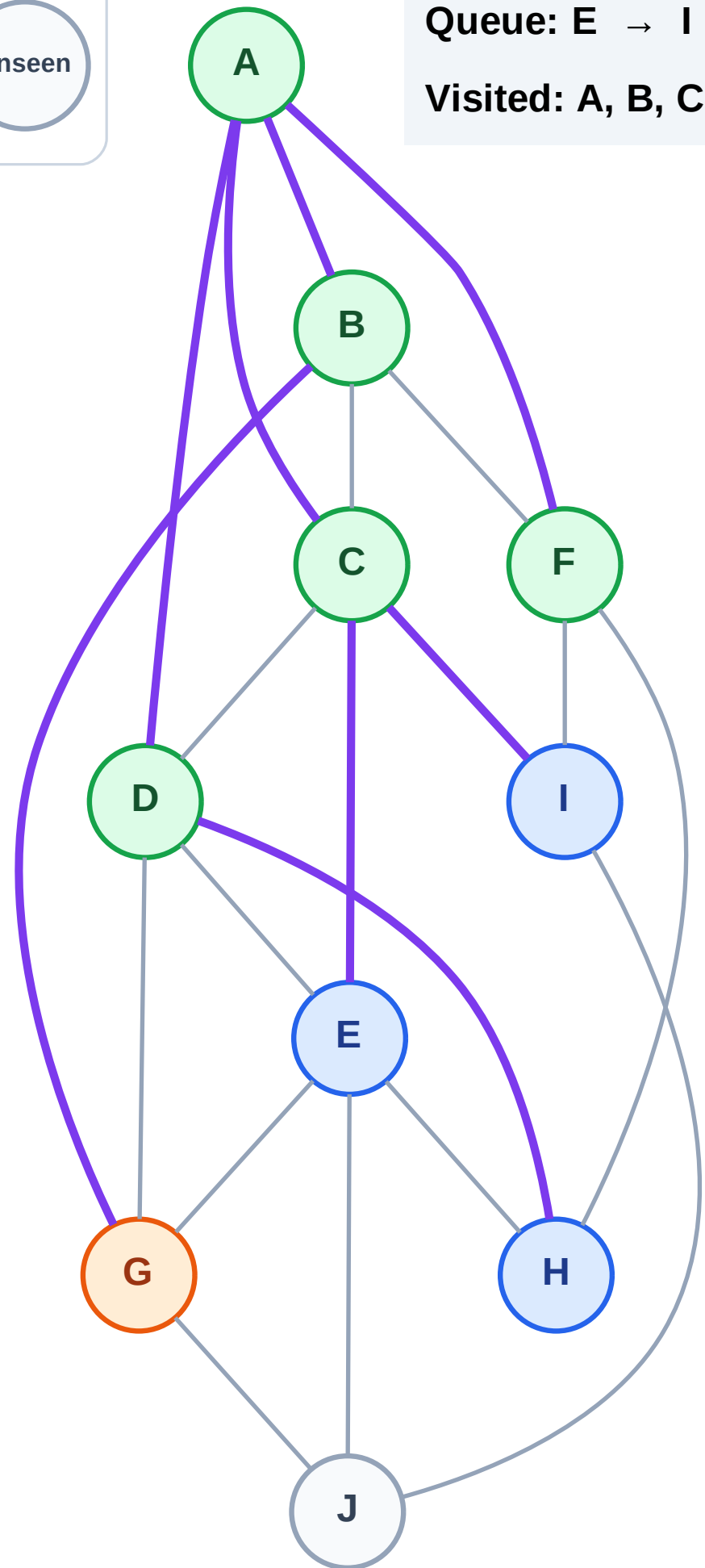
Step 12: Process node G

Legend



Queue: E → I → H

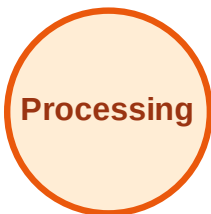
Visited: A, B, C, D, F



BFS Traversal

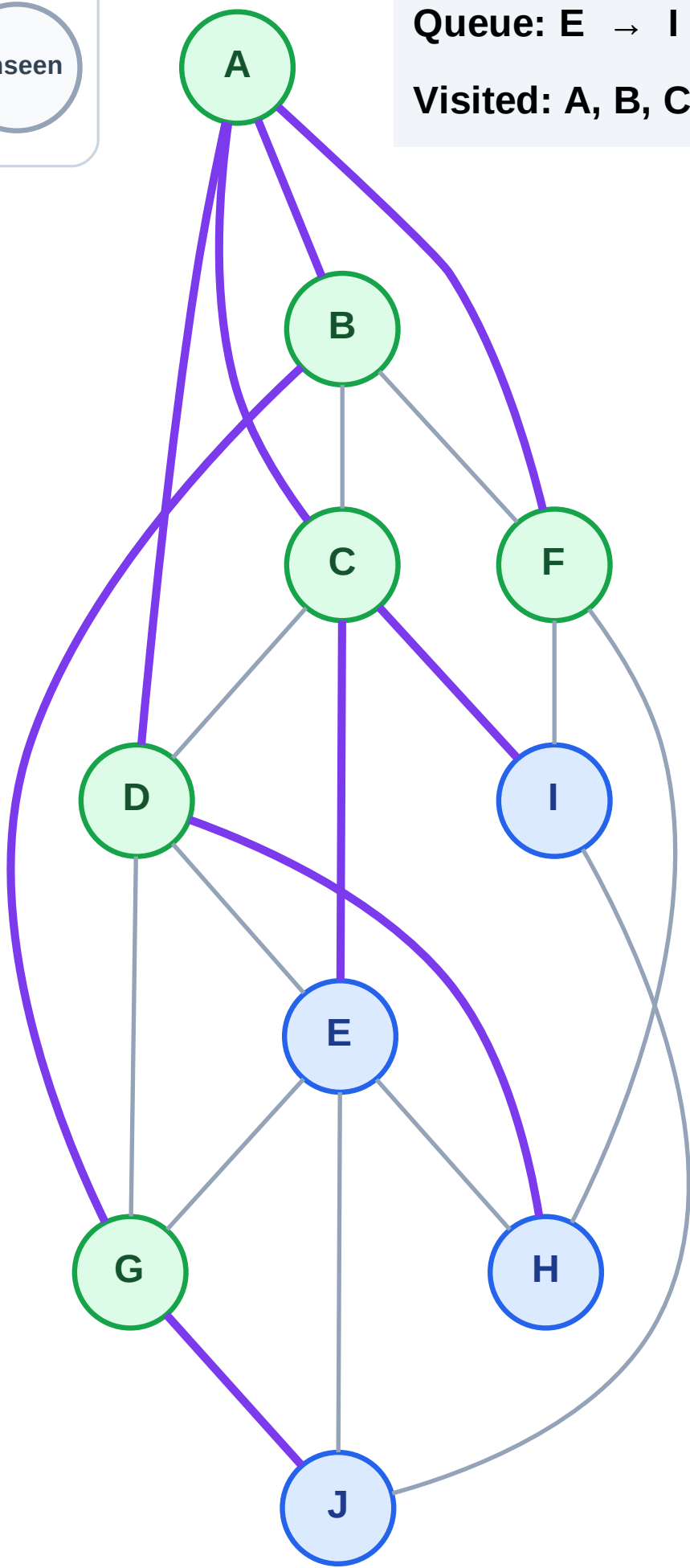
Step 13: Discover J from G

Legend



Queue: E → I → H → J

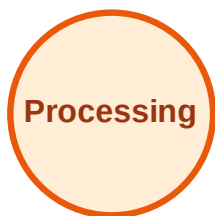
Visited: A, B, C, D, F, G



BFS Traversal

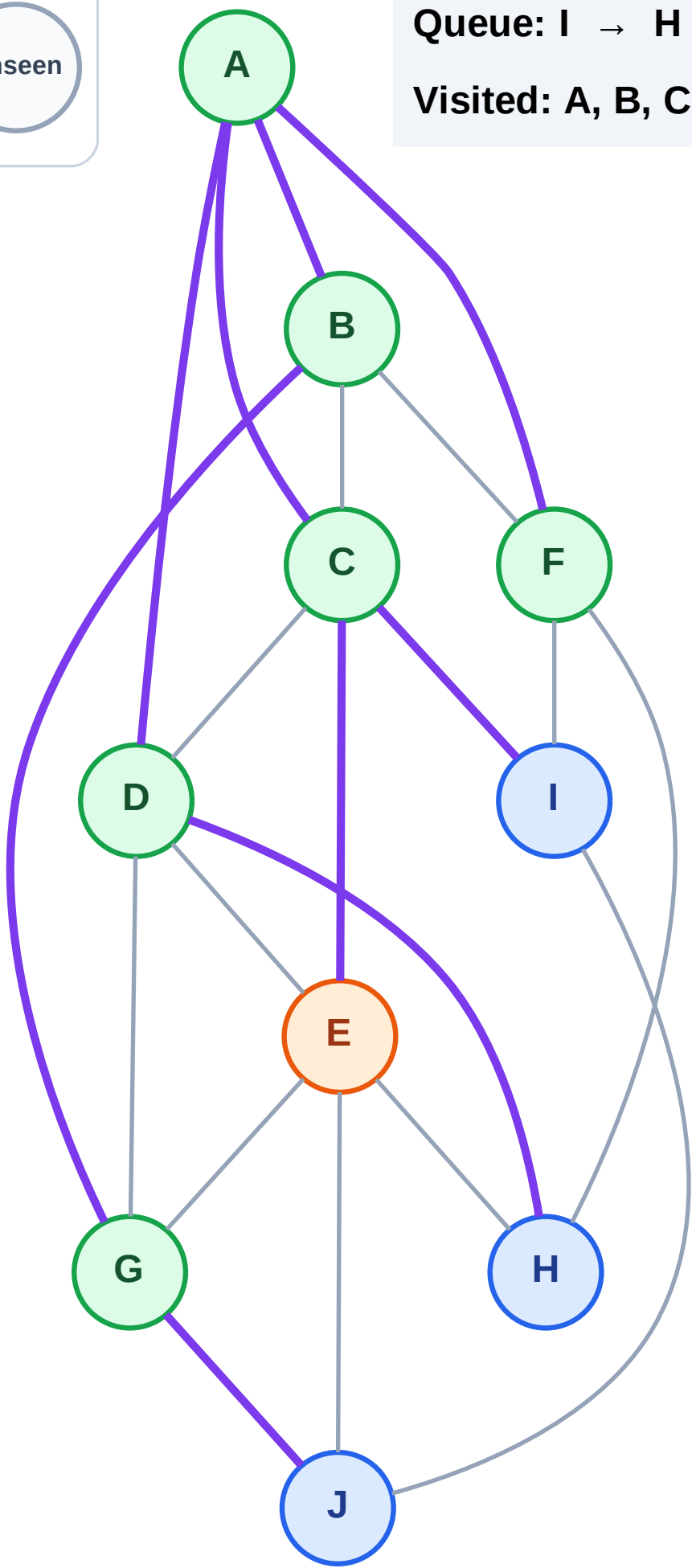
Step 14: Process node E

Legend



Queue: I → H → J

Visited: A, B, C, D, F, G



BFS Traversal

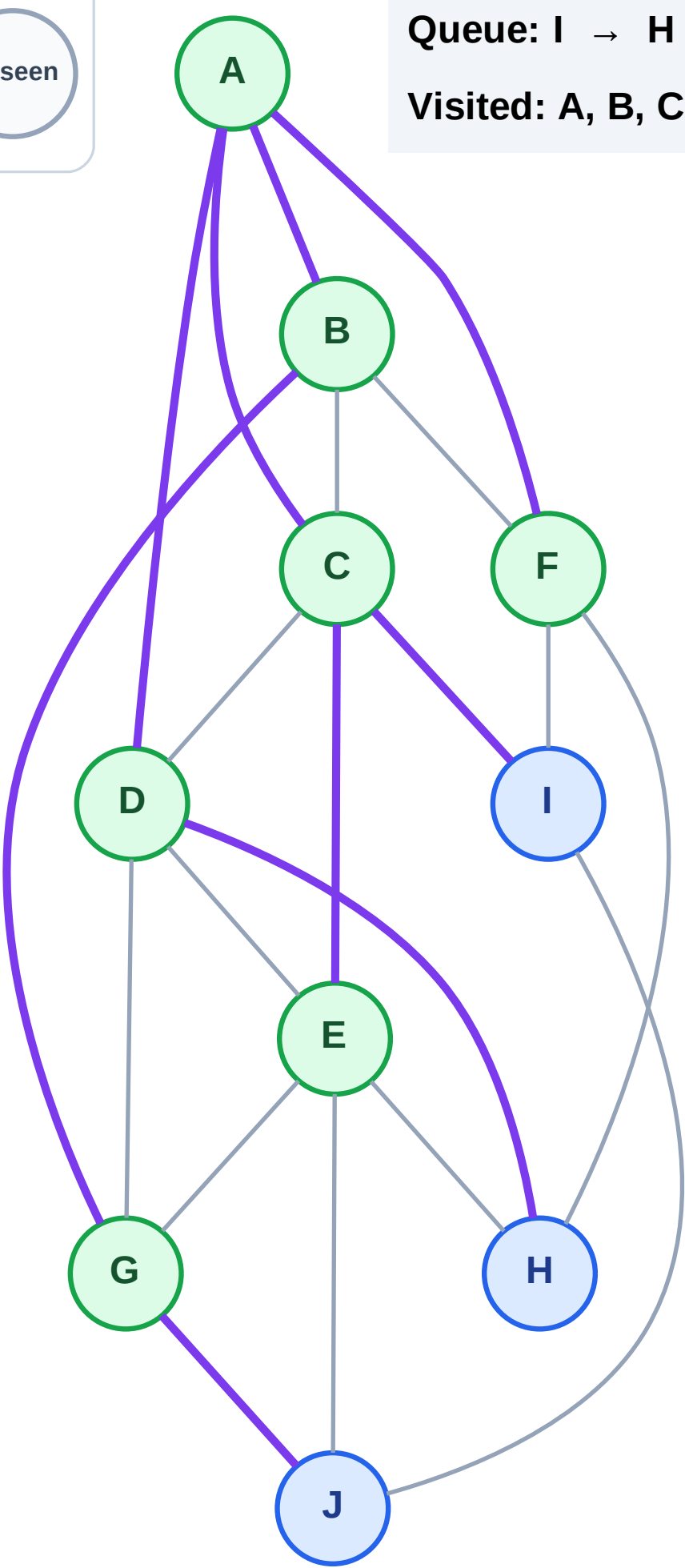
Step 15: No new neighbors from E

Legend



Queue: I → H → J

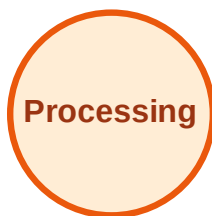
Visited: A, B, C, D, E, F, G



BFS Traversal

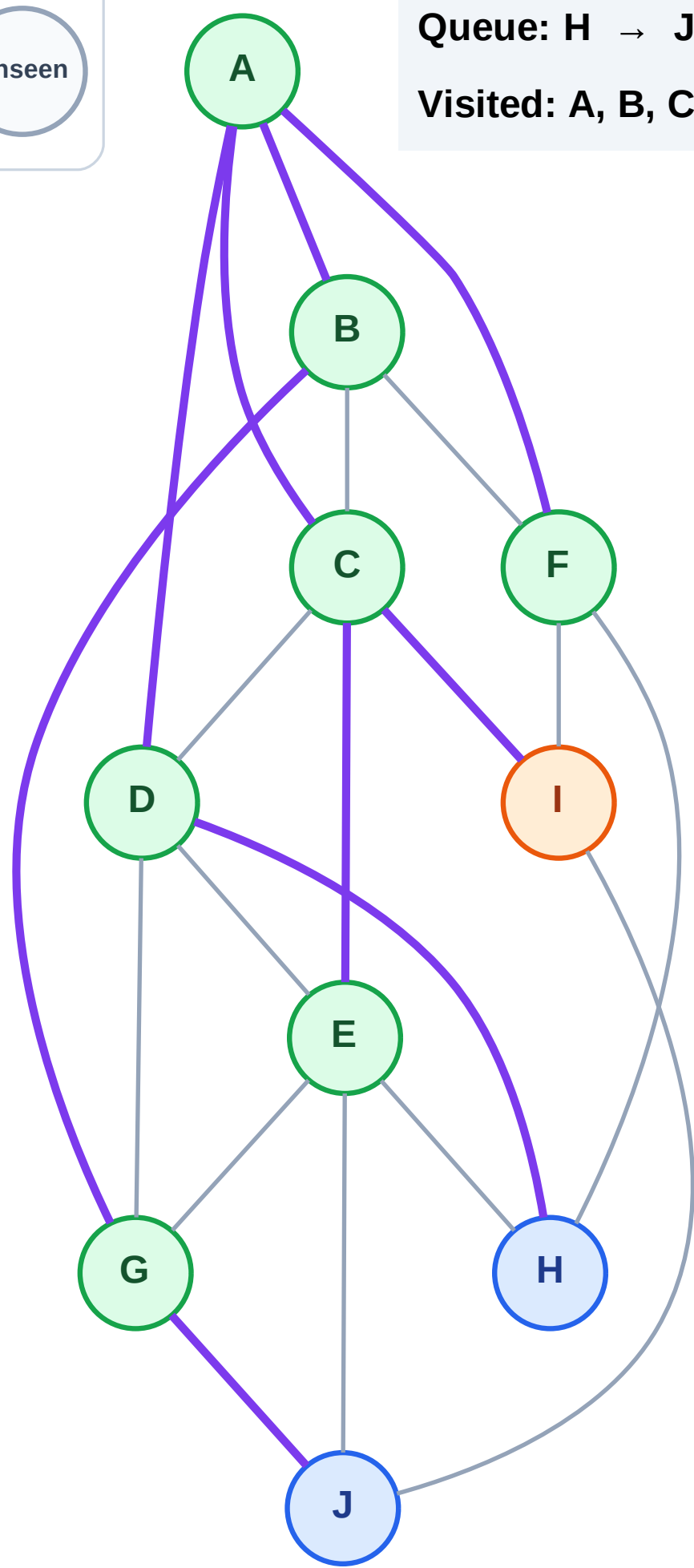
Step 16: Process node I

Legend



Queue: H → J

Visited: A, B, C, D, E, F, G



BFS Traversal

Step 17: No new neighbors from I

Legend

Complete

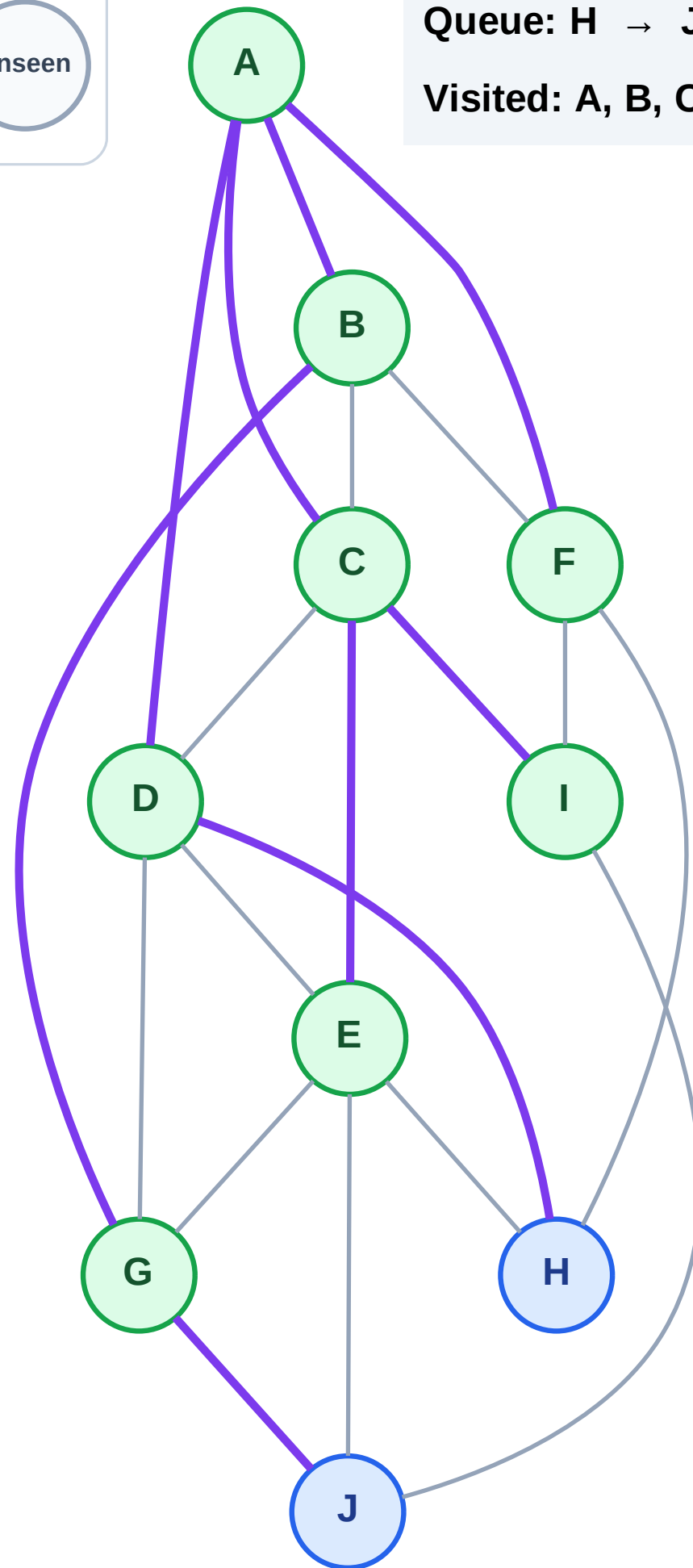
Processing

Discovered

Unseen

Queue: H → J

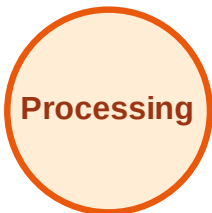
Visited: A, B, C, D, E, F, G, I



BFS Traversal

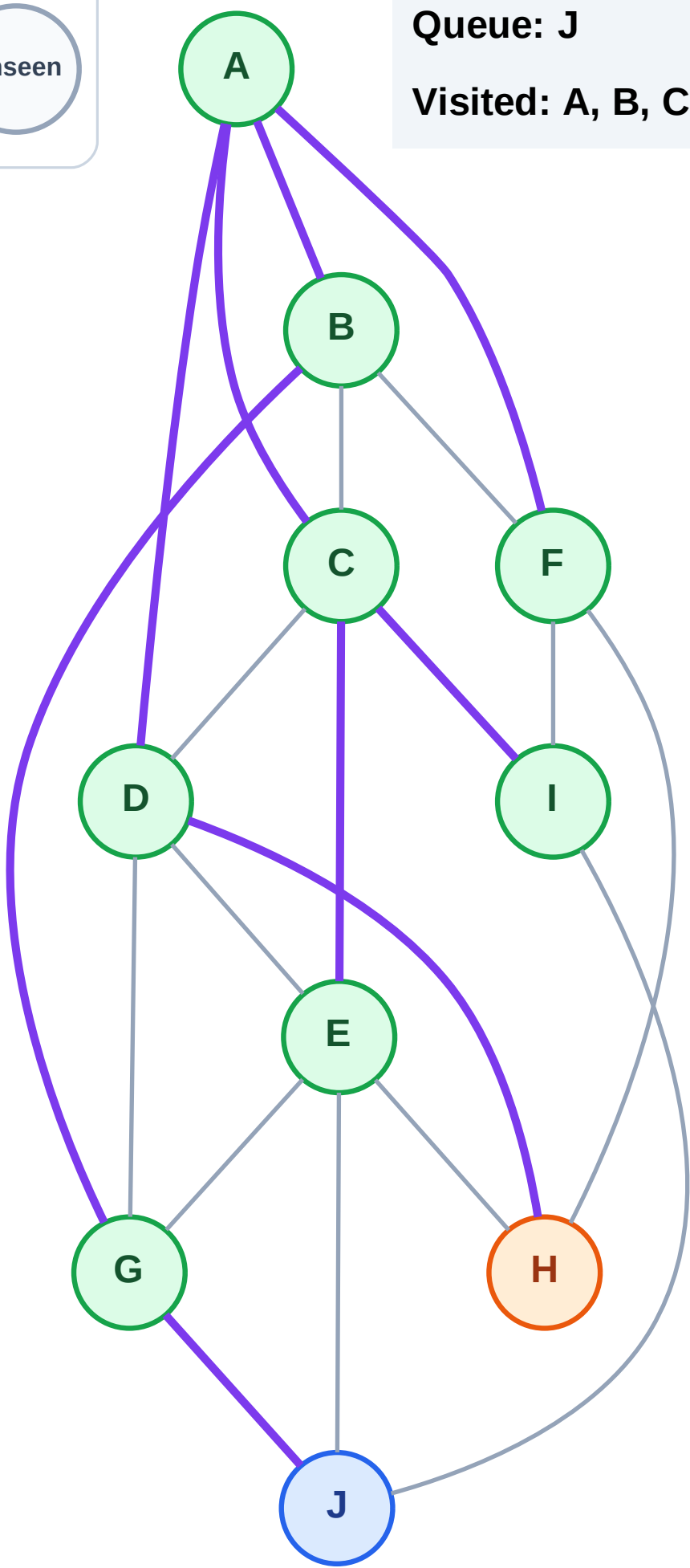
Step 18: Process node H

Legend



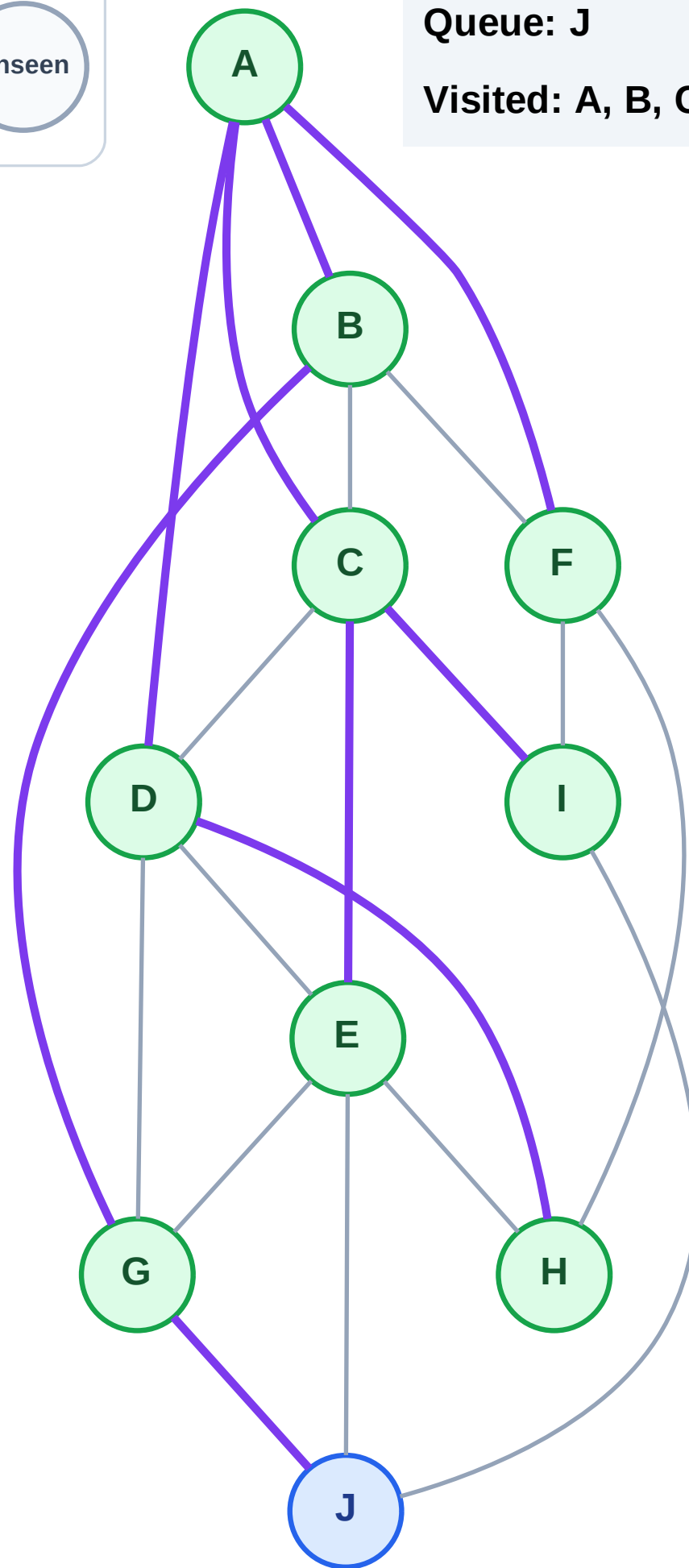
Queue: J

Visited: A, B, C, D, E, F, G, I



Step 19: No new neighbors from H

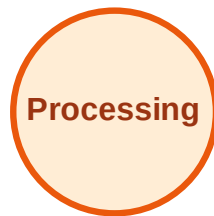
Step 19: No new neighbors from H



BFS Traversal

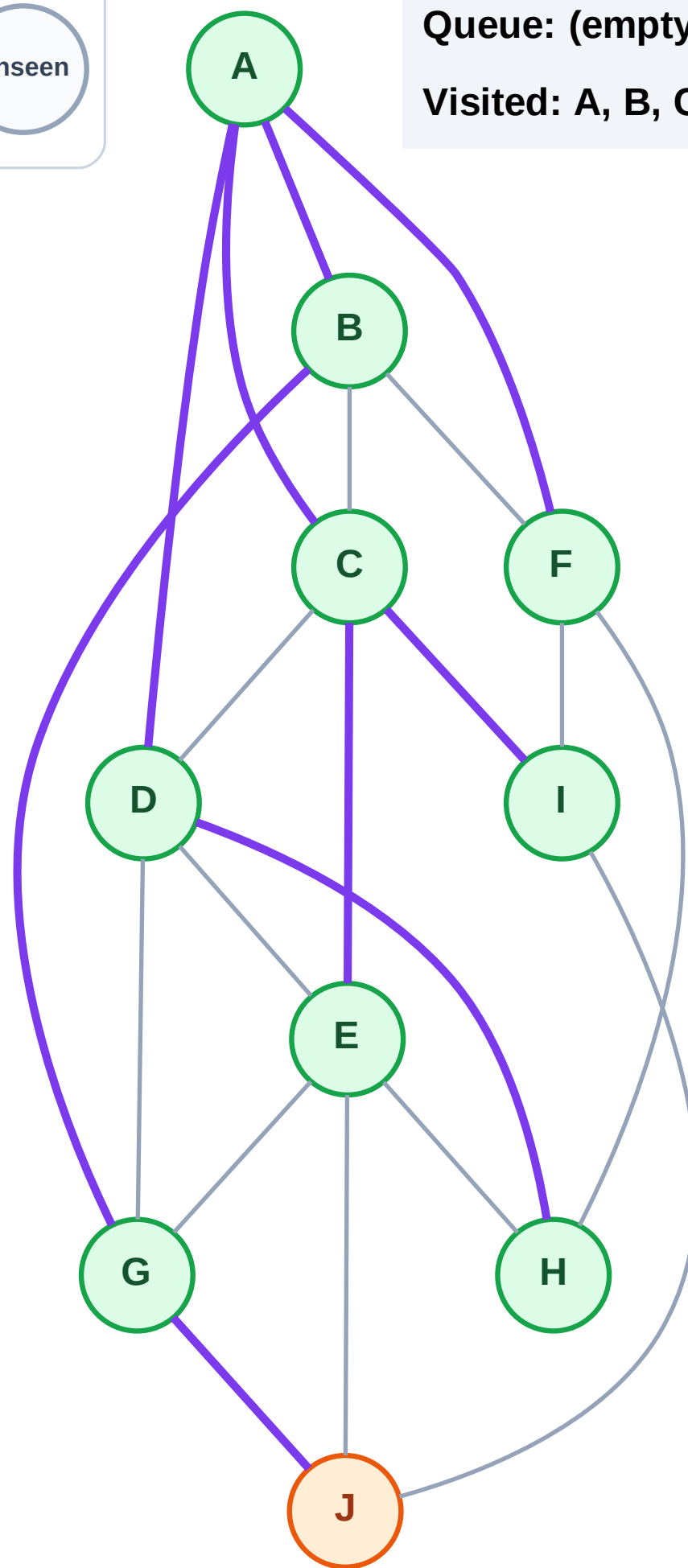
Step 20: Process node J

Legend



Queue: (empty)

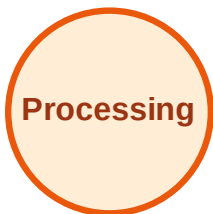
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

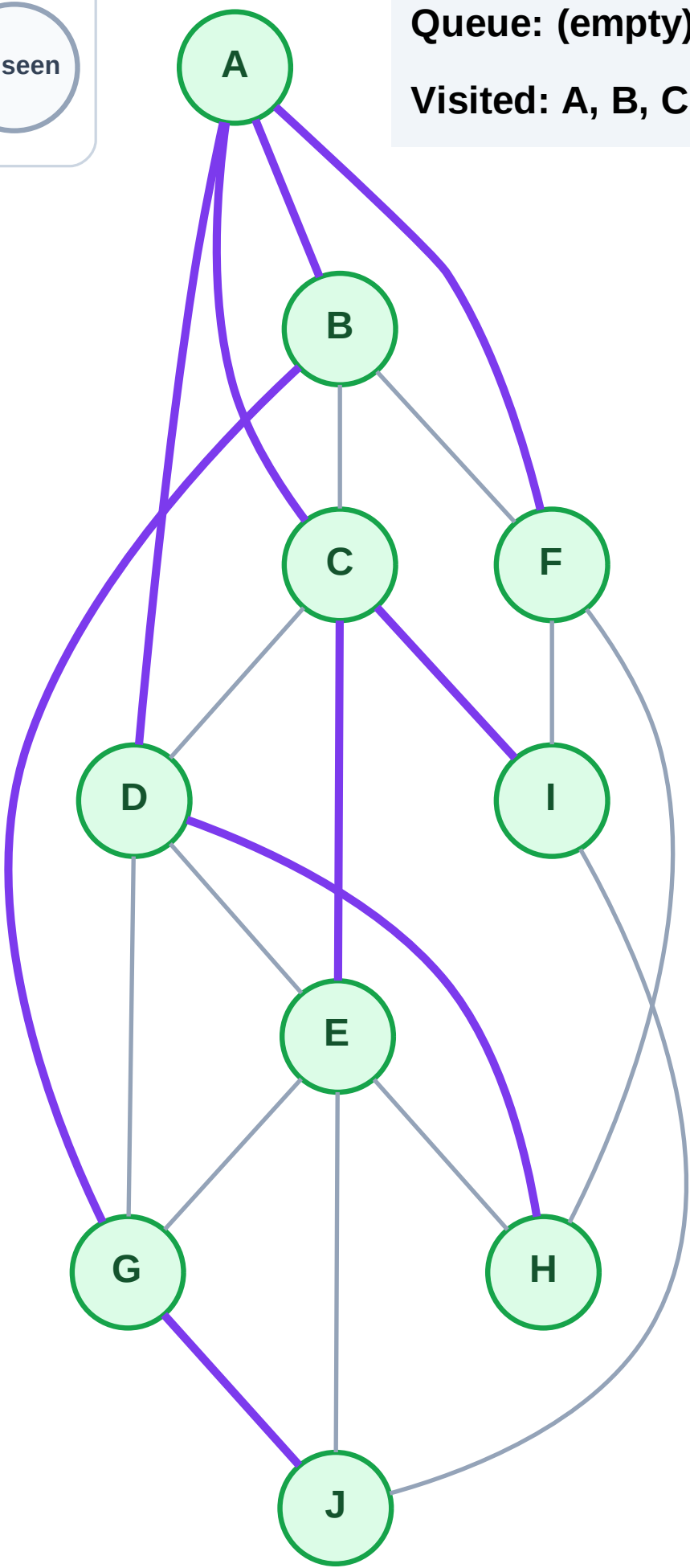
Step 21: No new neighbors from J

Legend



Queue: (empty)

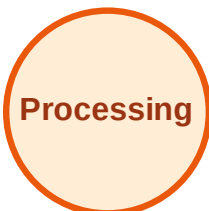
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

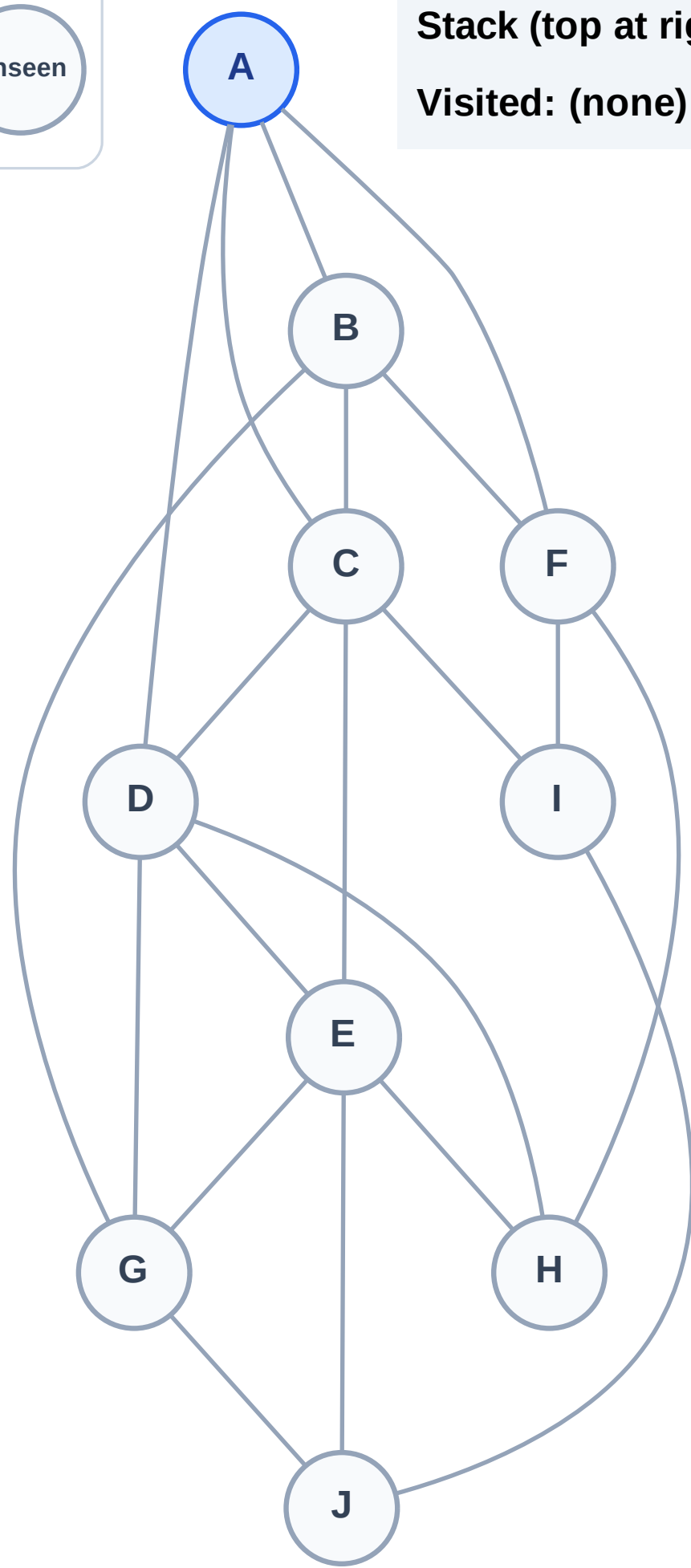
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

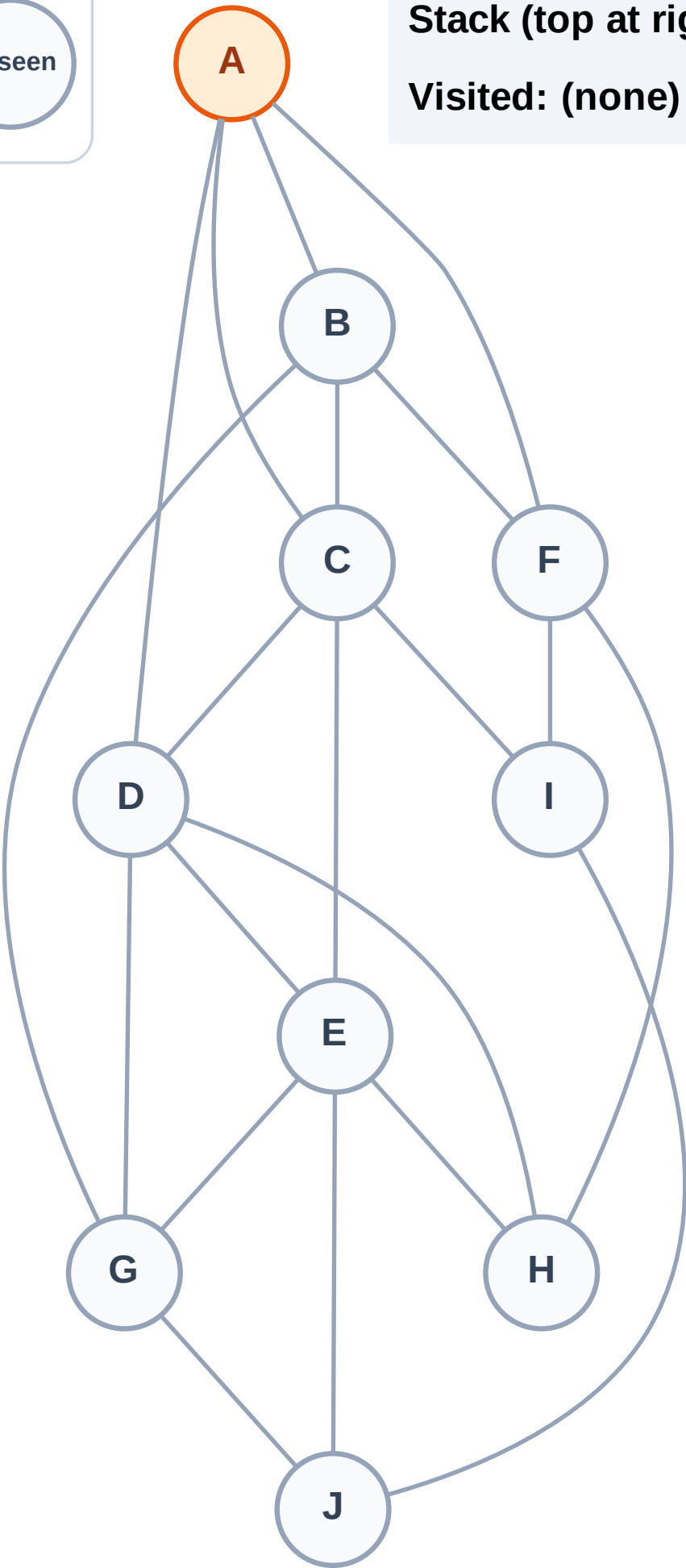
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover F, D, C, B from A

Legend

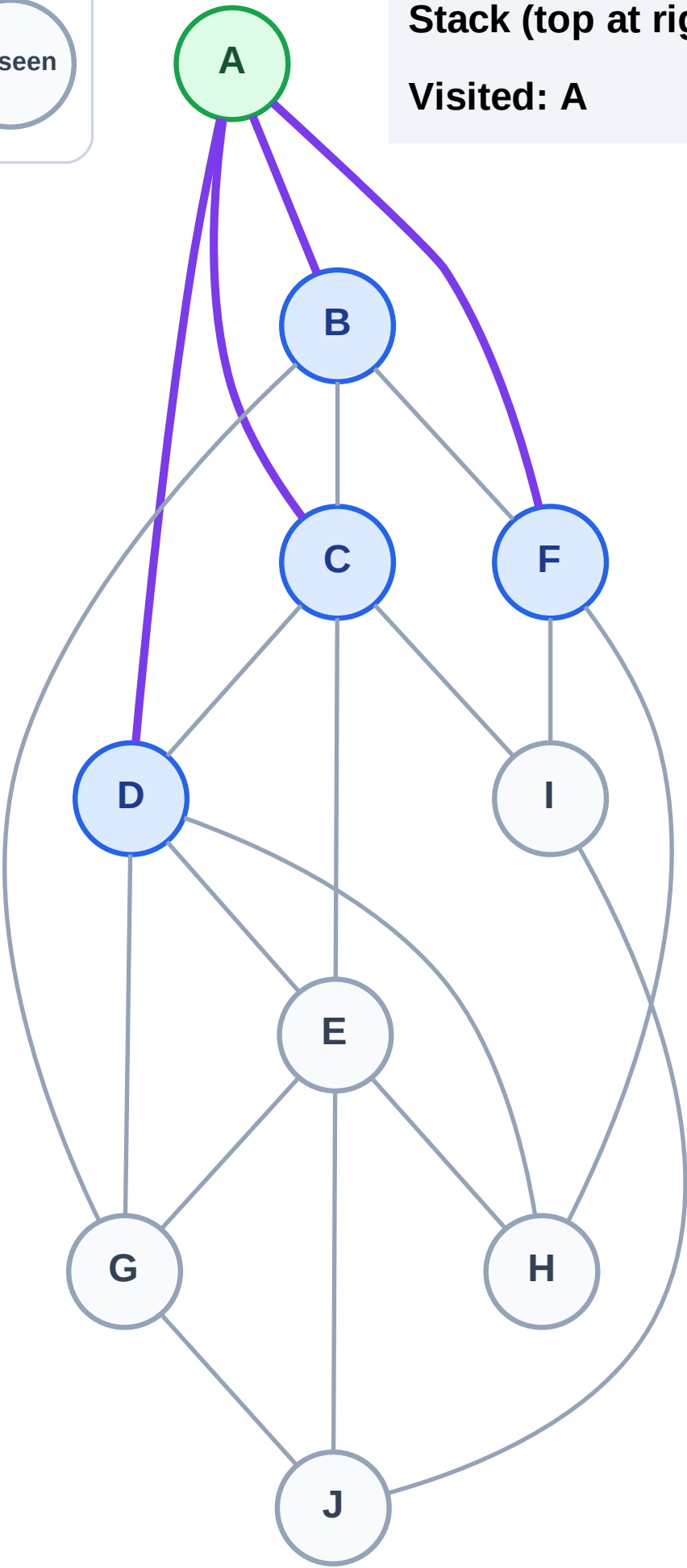
Complete

Processing

Discovered

Unseen

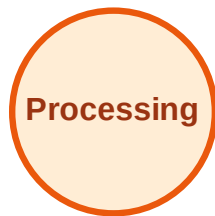
Stack (top at right): F | D | C | B
Visited: A



DFS Traversal

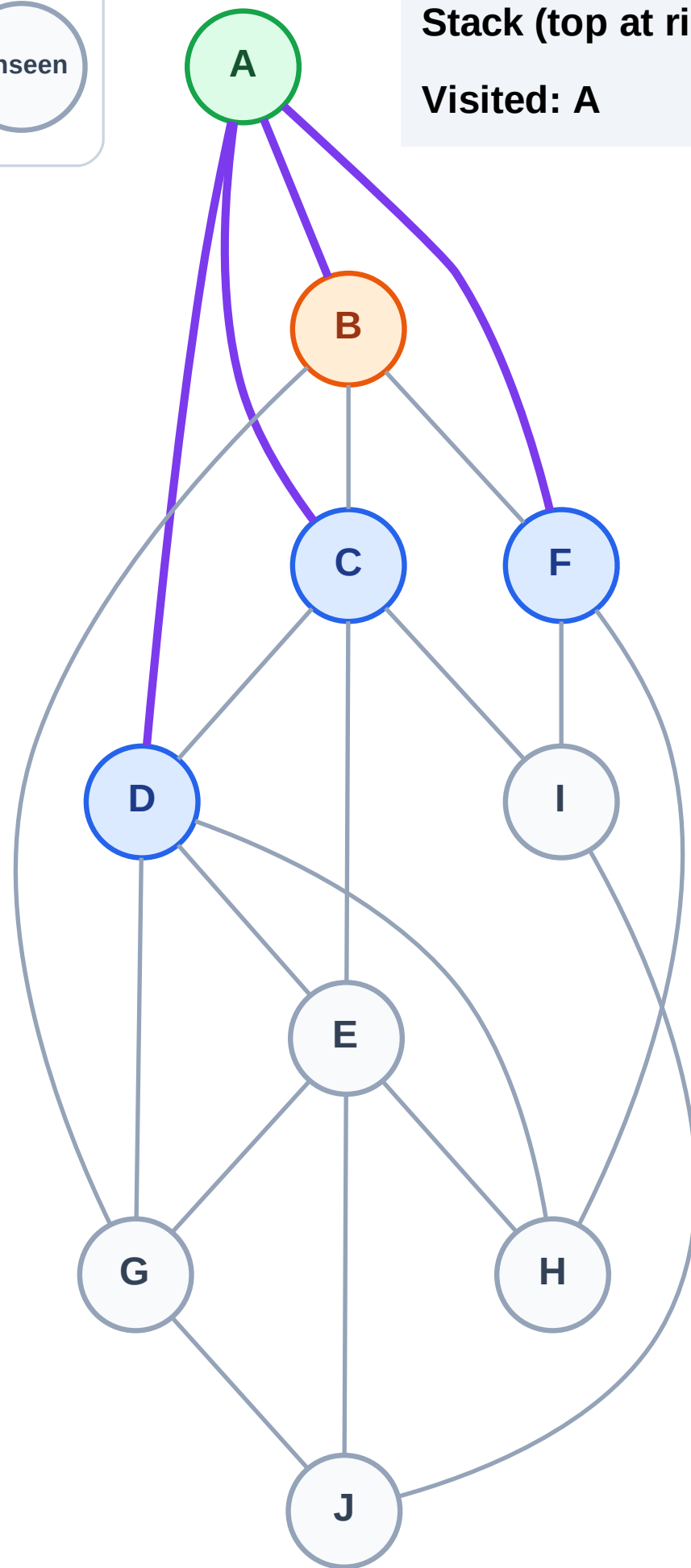
Step 4: Process node B

Legend



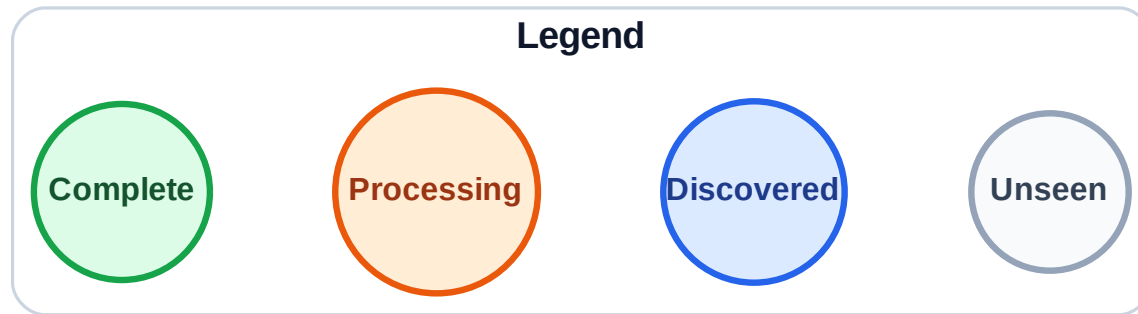
Stack (top at right): F | D | C

Visited: A

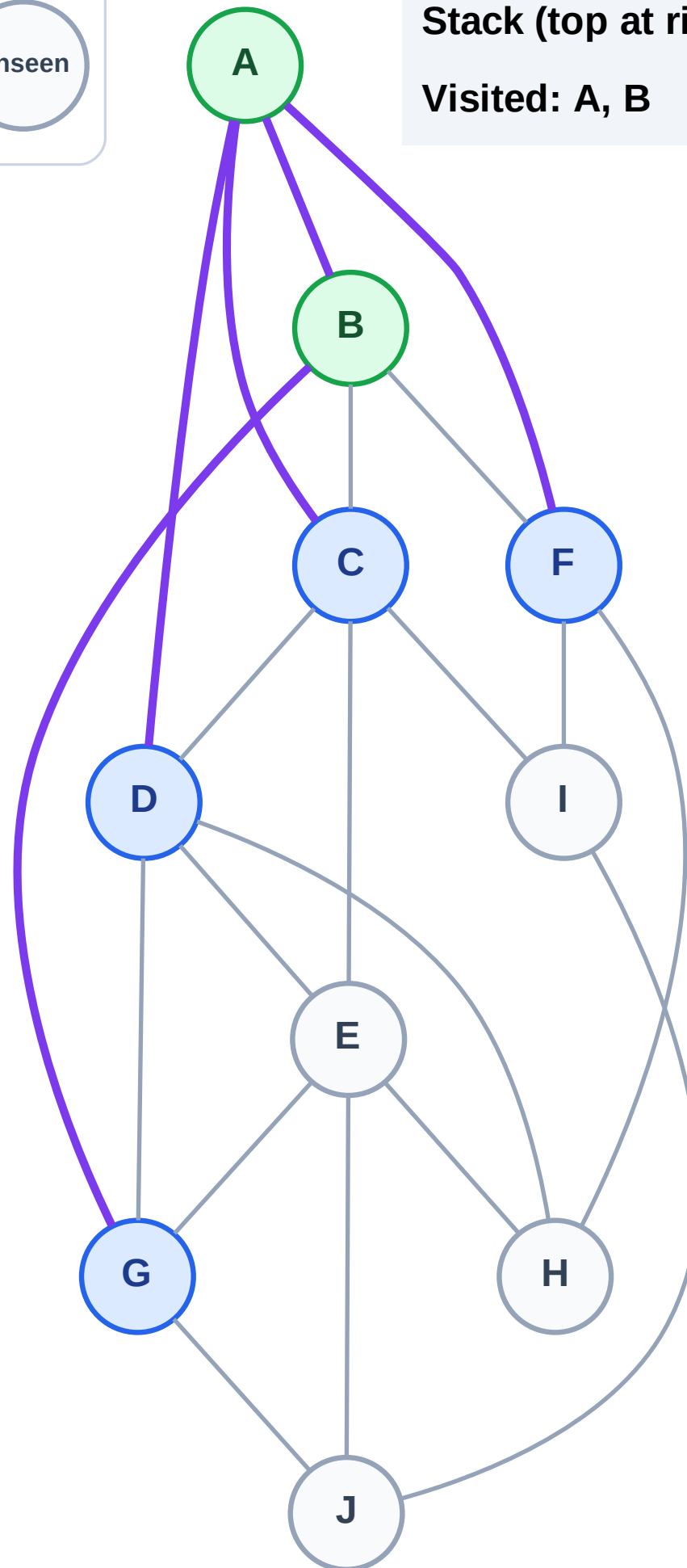


DFS Traversal

Step 5: Discover G from B



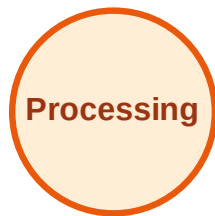
Stack (top at right): F | D | C | G
Visited: A, B



DFS Traversal

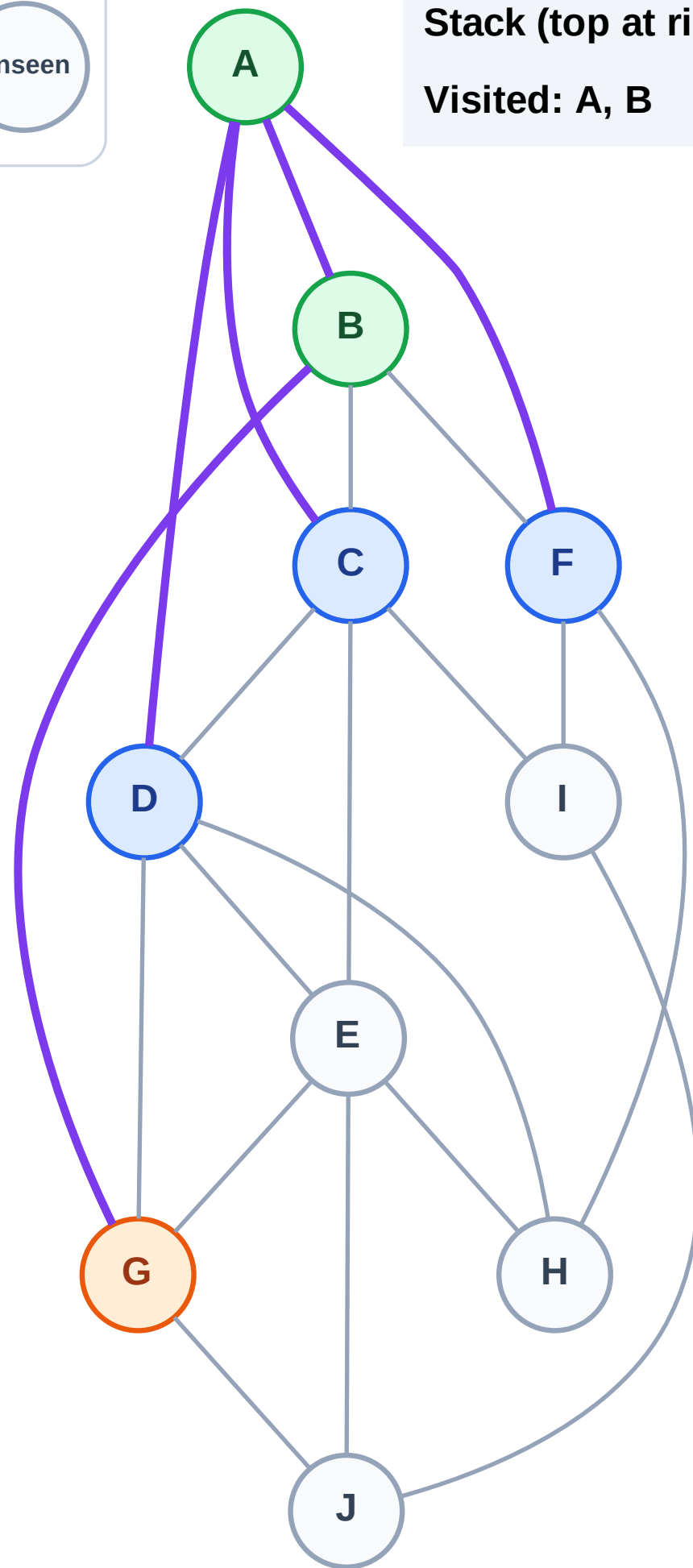
Step 6: Process node G

Legend



Stack (top at right): F | D | C

Visited: A, B



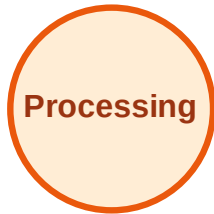
DFS Traversal

Step 7: Discover J, E from G

Legend



Complete



Processing



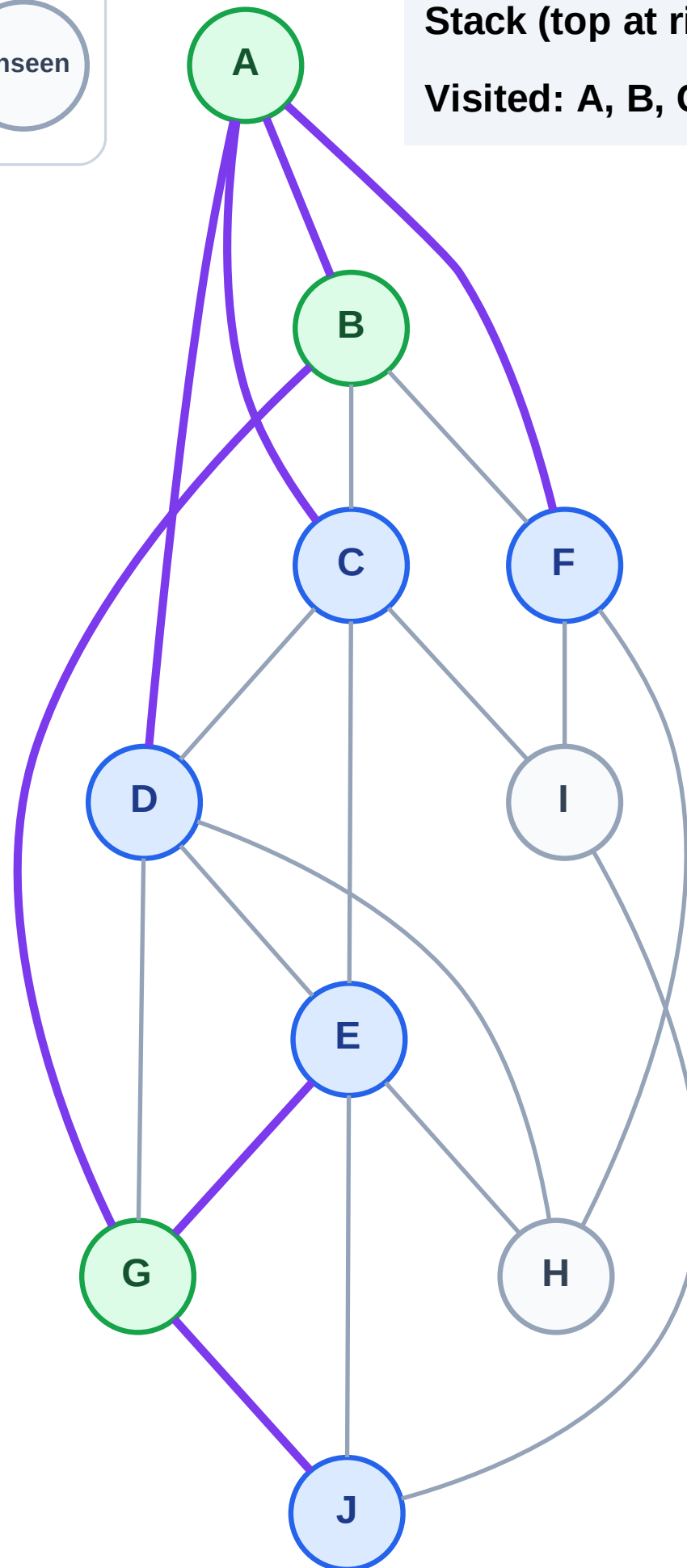
Discovered



Unseen

Stack (top at right): F | D | C | J | E

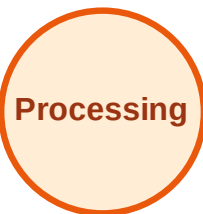
Visited: A, B, G



DFS Traversal

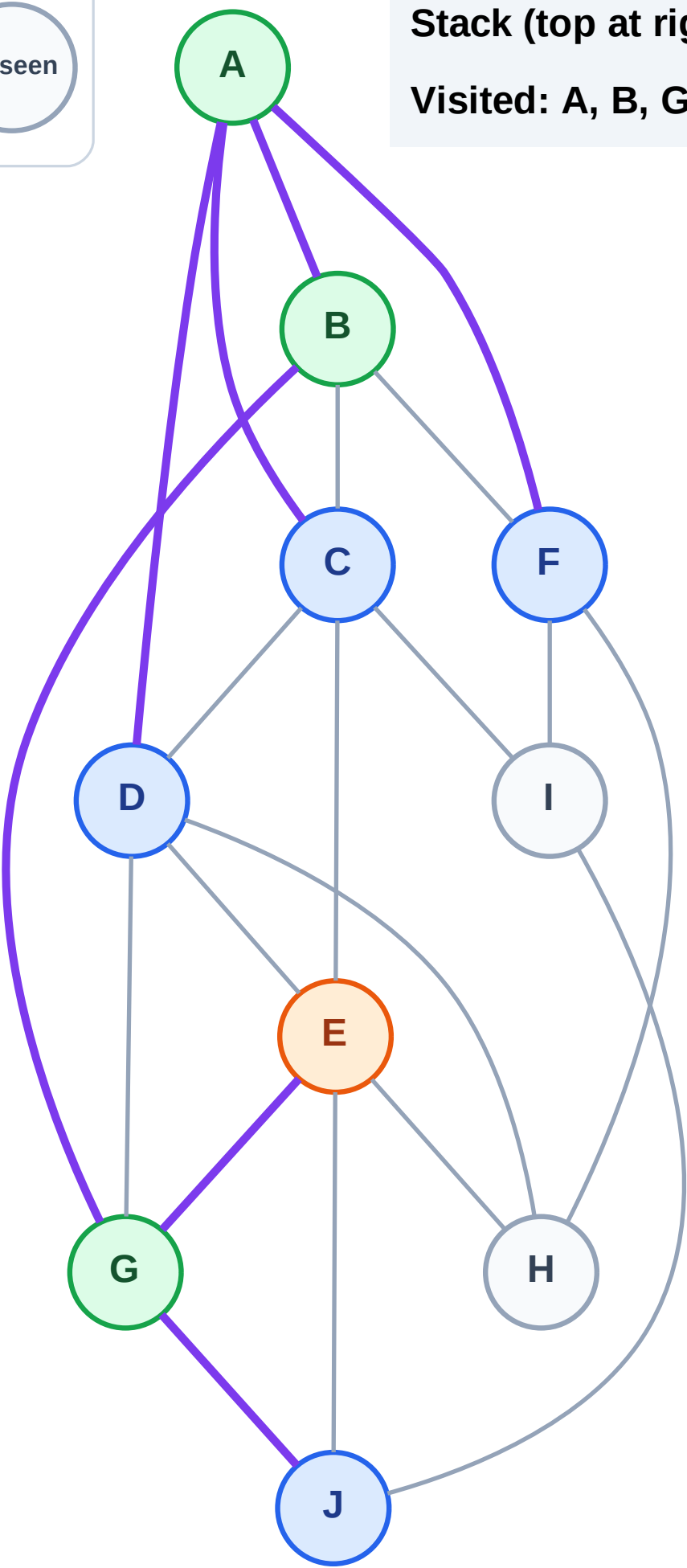
Step 8: Process node E

Legend



Stack (top at right): F | D | C | J

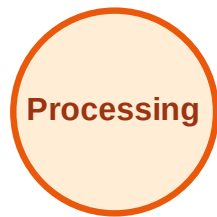
Visited: A, B, G



DFS Traversal

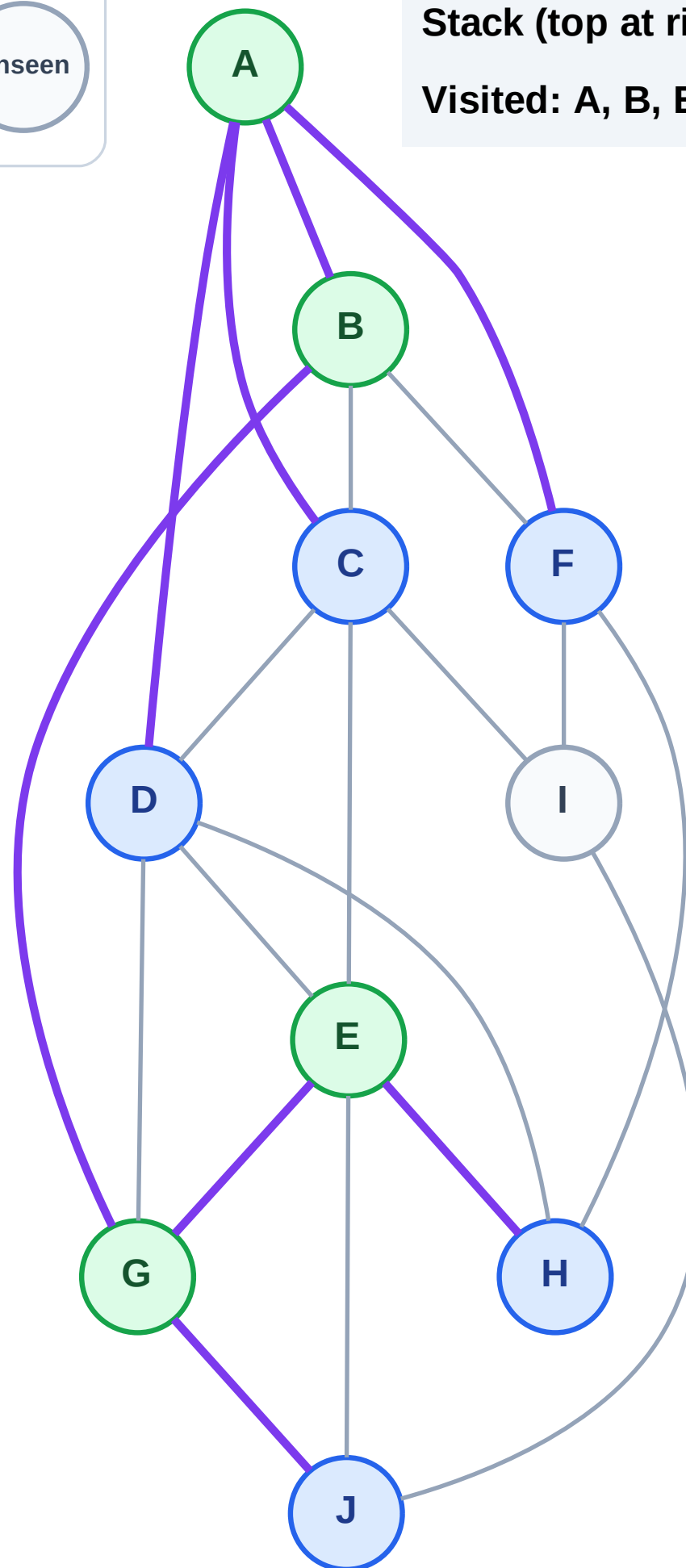
Step 9: Discover H from E

Legend



Stack (top at right): F | D | C | J | H

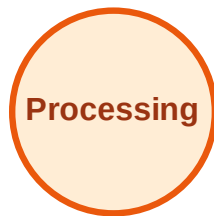
Visited: A, B, E, G



DFS Traversal

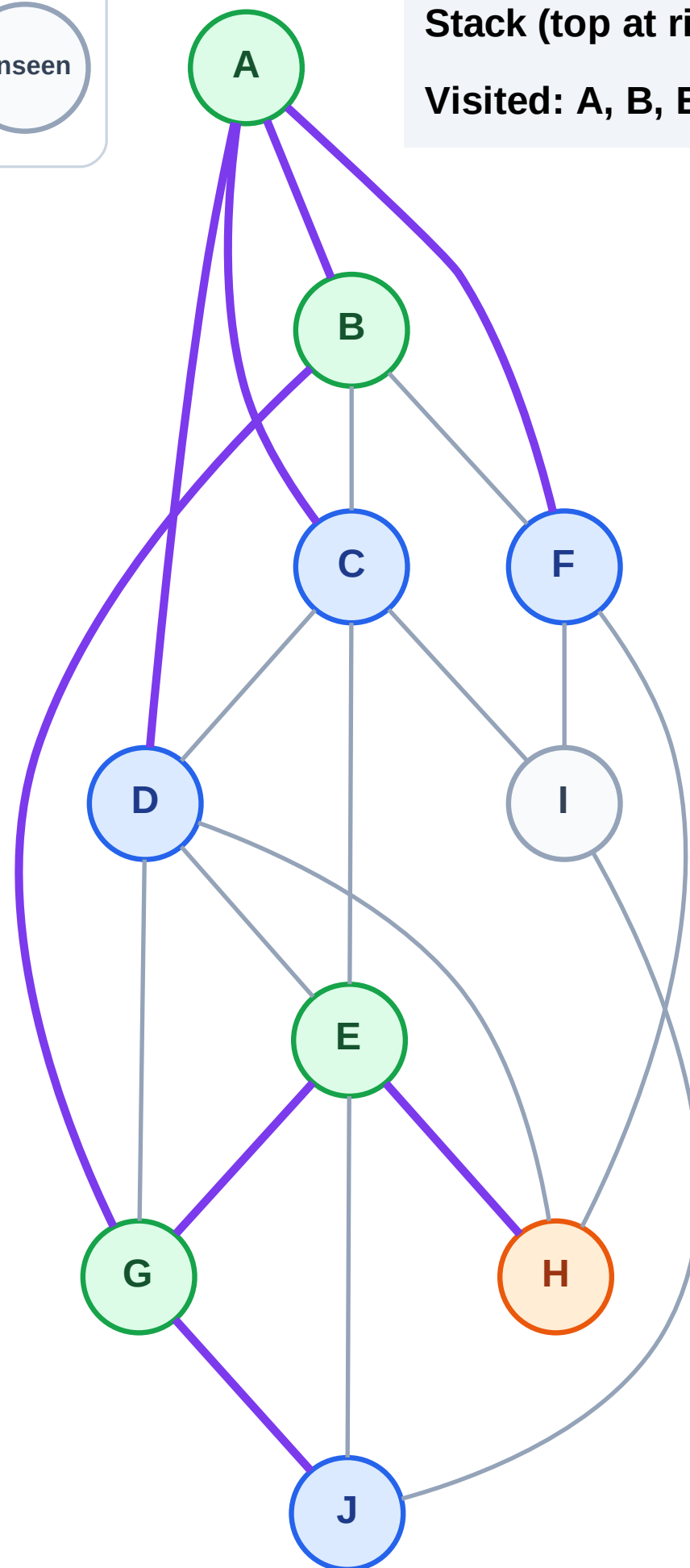
Step 10: Process node H

Legend



Stack (top at right): F | D | C | J

Visited: A, B, E, G



DFS Traversal

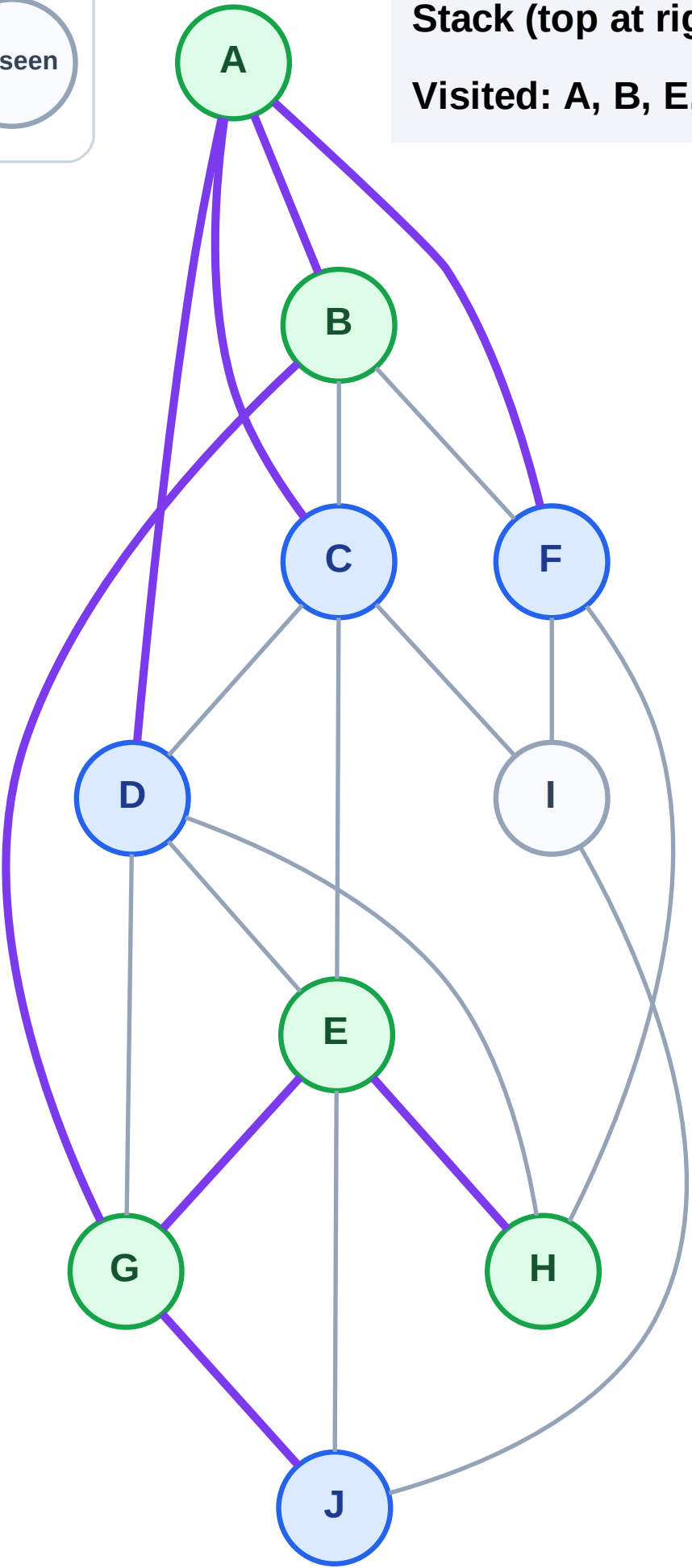
Step 11: No new neighbors from H

Legend



Stack (top at right): F | D | C | J

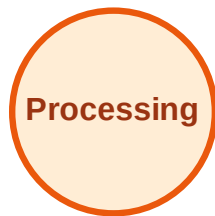
Visited: A, B, E, G, H



DFS Traversal

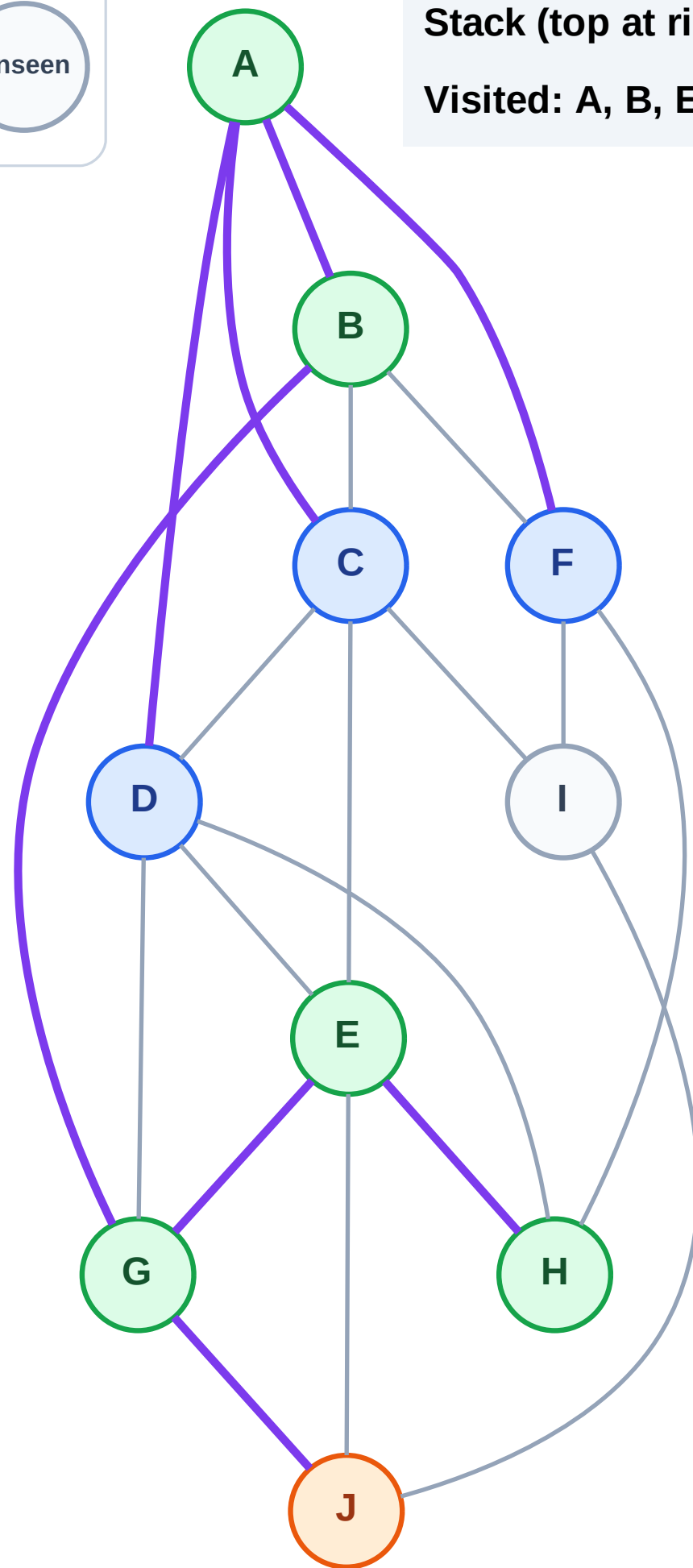
Step 12: Process node J

Legend



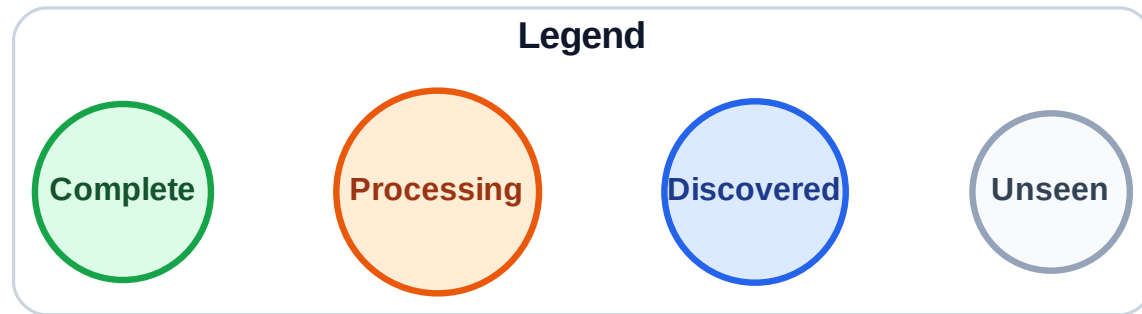
Stack (top at right): F | D | C

Visited: A, B, E, G, H

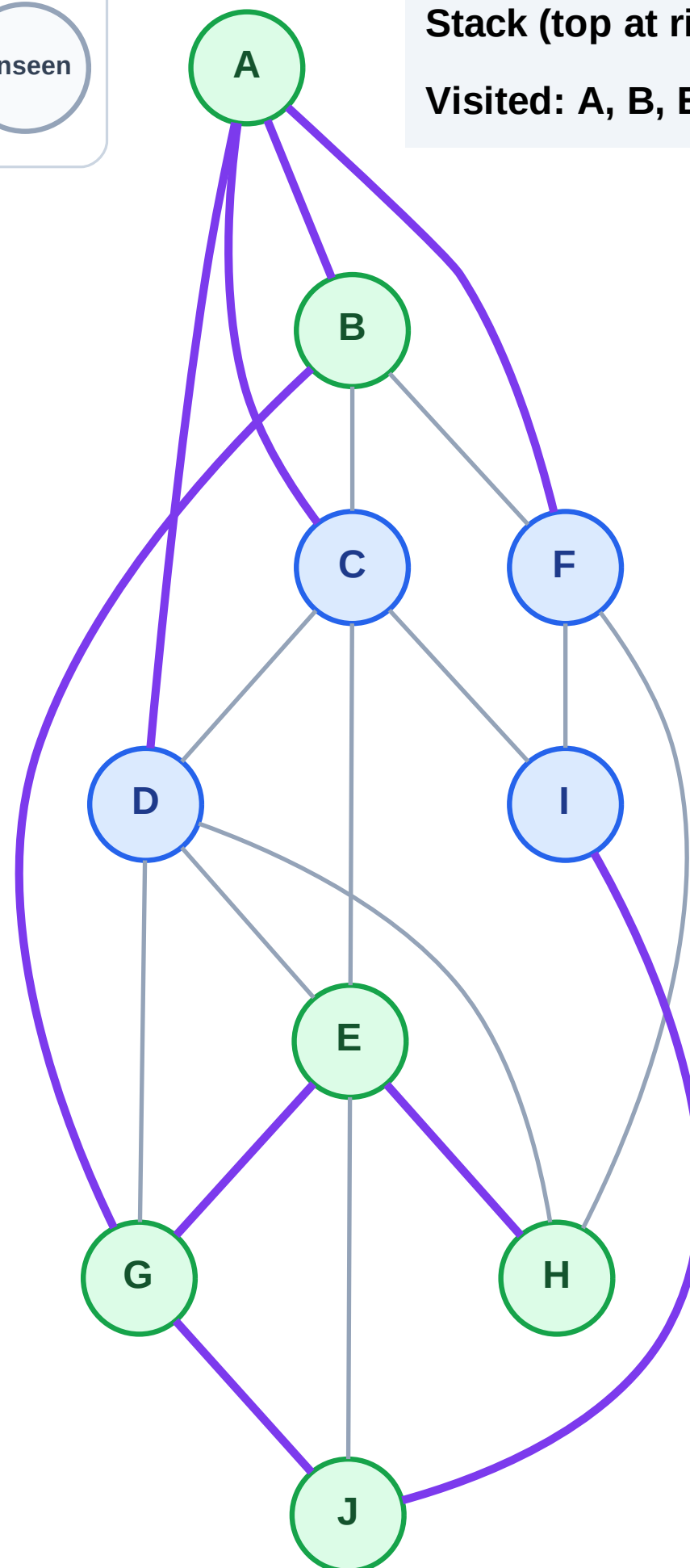


DFS Traversal

Step 13: Discover I from J



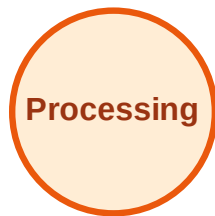
Stack (top at right): F | D | C | I
Visited: A, B, E, G, H, J



DFS Traversal

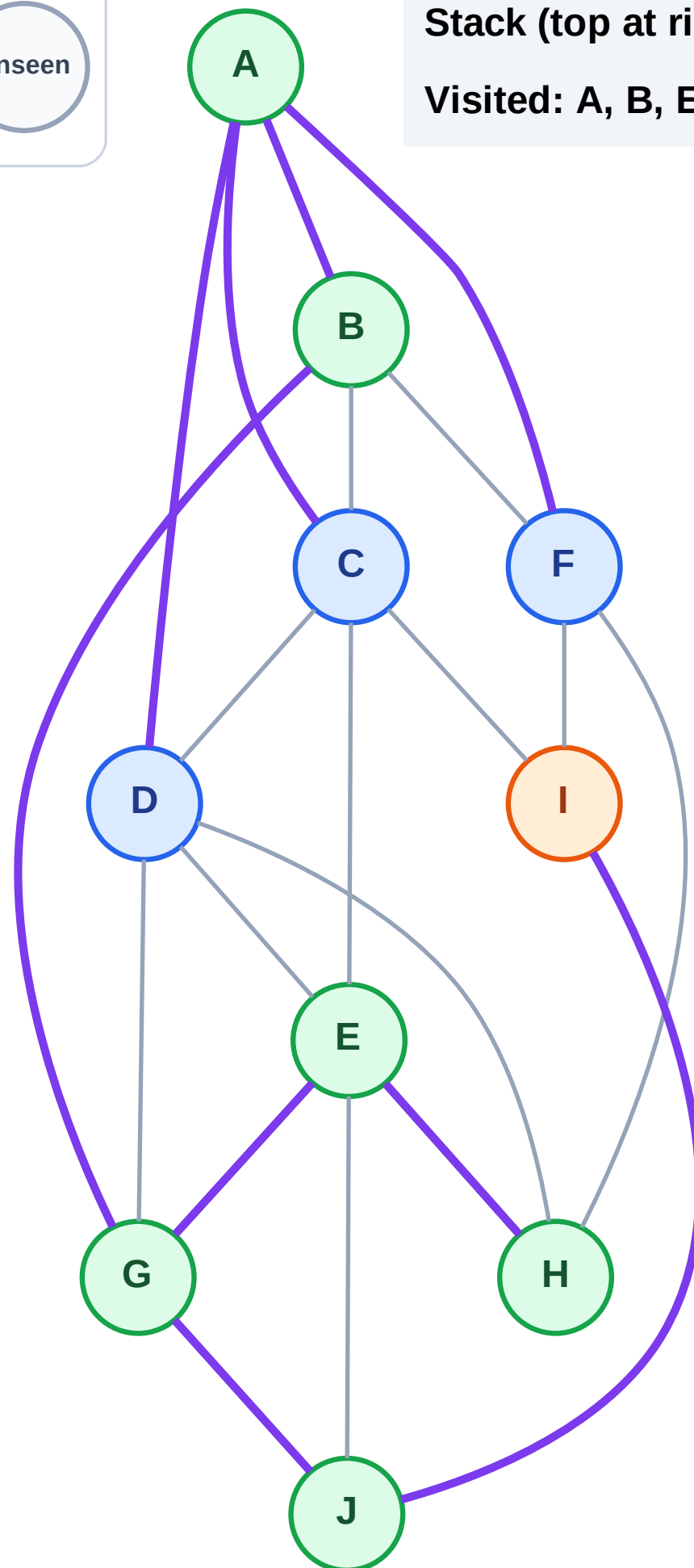
Step 14: Process node I

Legend



Stack (top at right): F | D | C

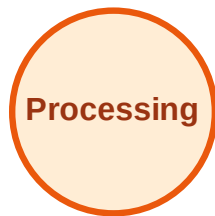
Visited: A, B, E, G, H, J



DFS Traversal

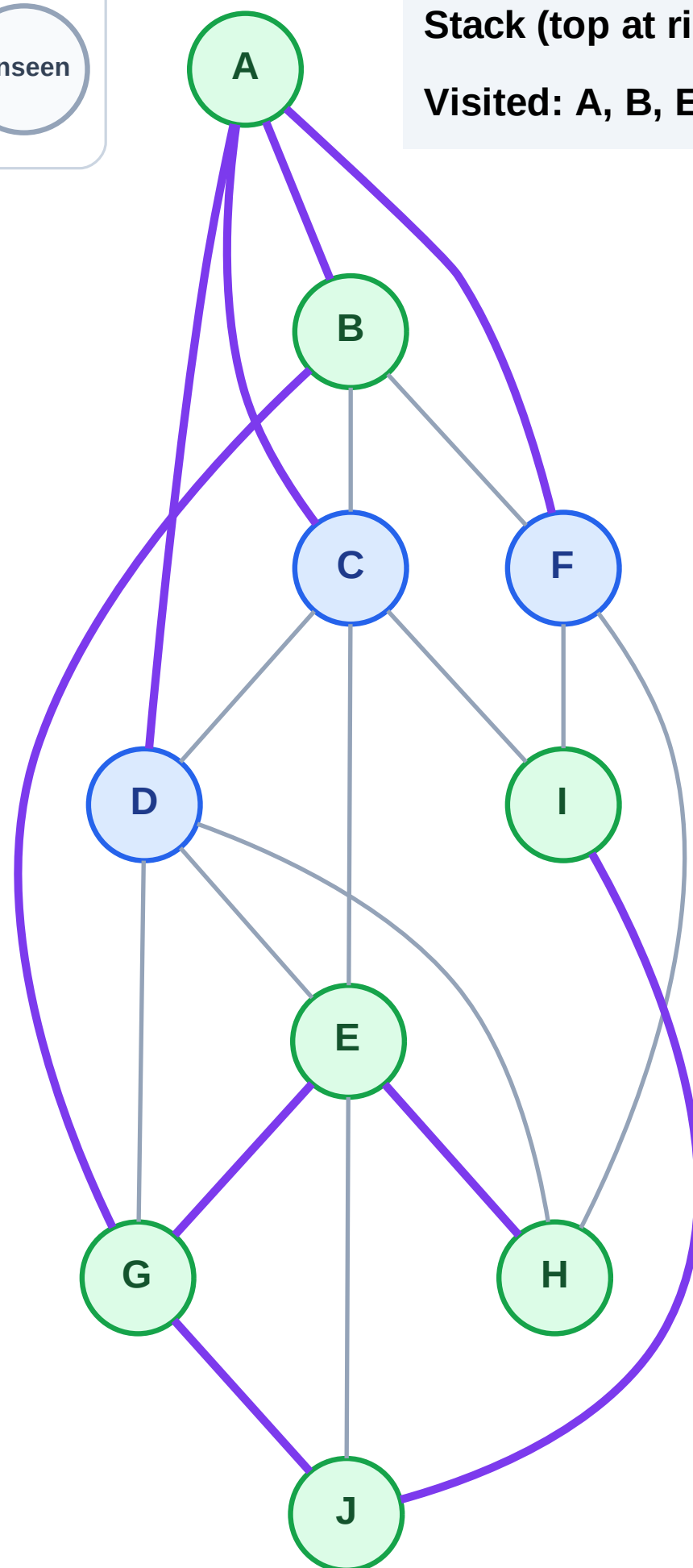
Step 15: No new neighbors from I

Legend



Stack (top at right): F | D | C

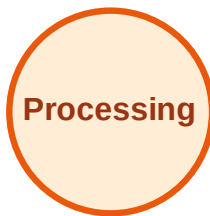
Visited: A, B, E, G, H, I, J



DFS Traversal

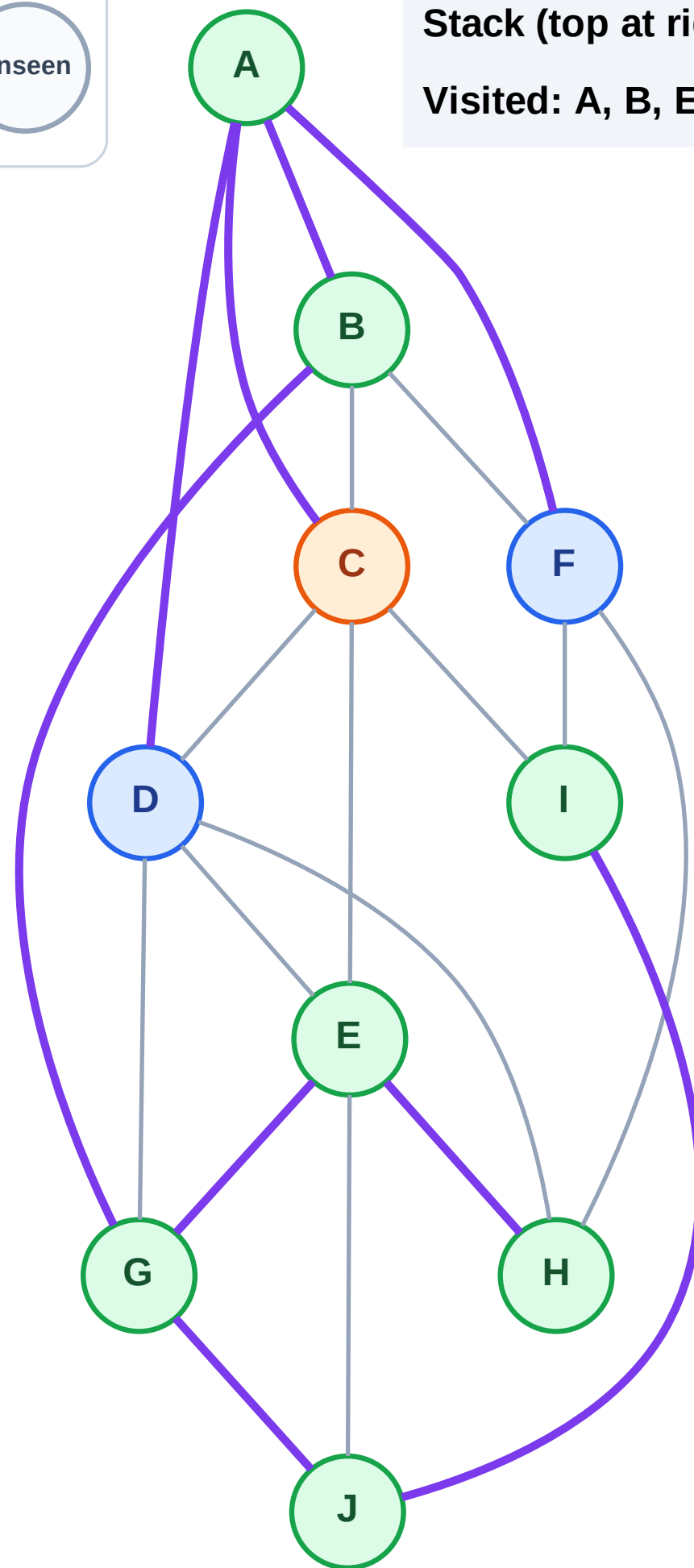
Step 16: Process node C

Legend



Stack (top at right): F | D

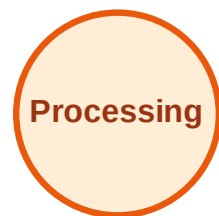
Visited: A, B, E, G, H, I, J



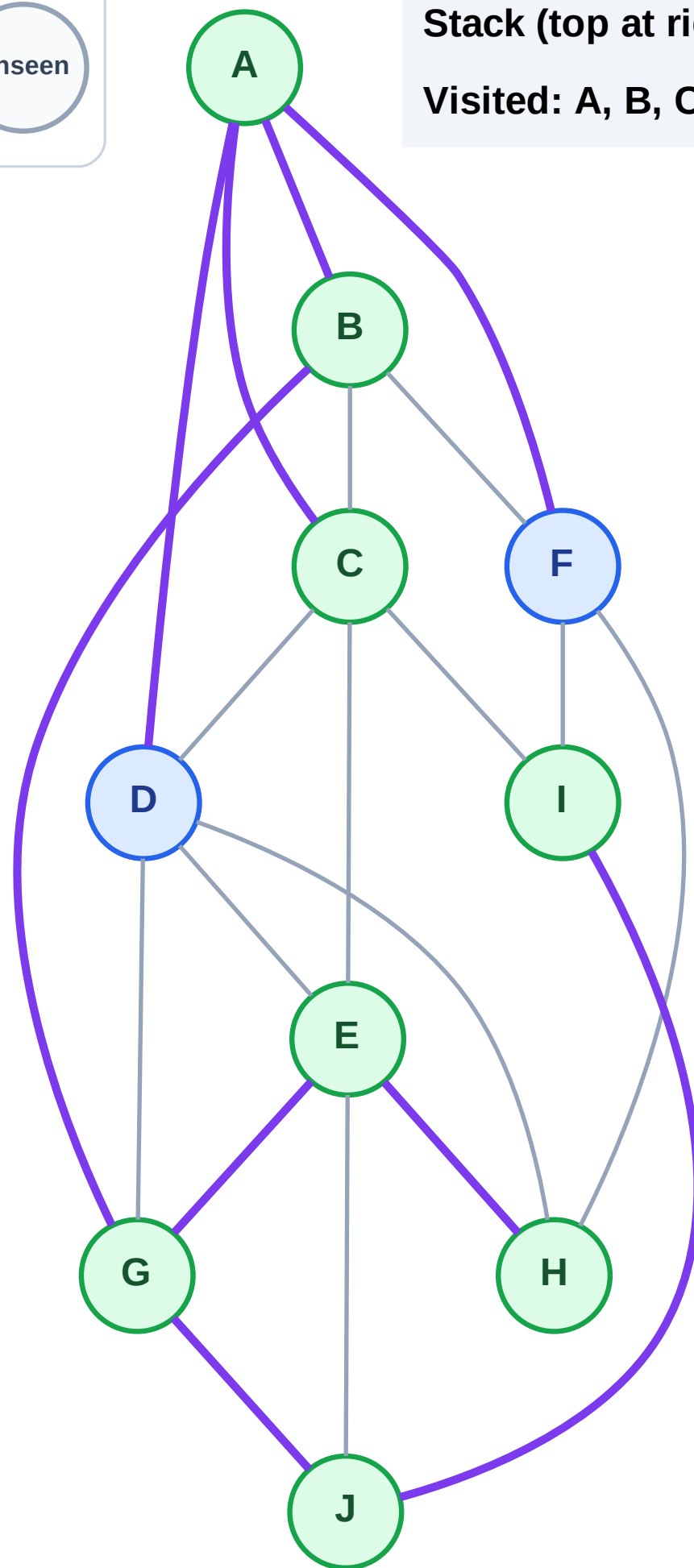
DFS Traversal

Step 17: No new neighbors from C

Legend



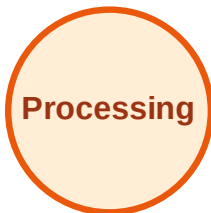
Stack (top at right): F | D
Visited: A, B, C, E, G, H, I, J



DFS Traversal

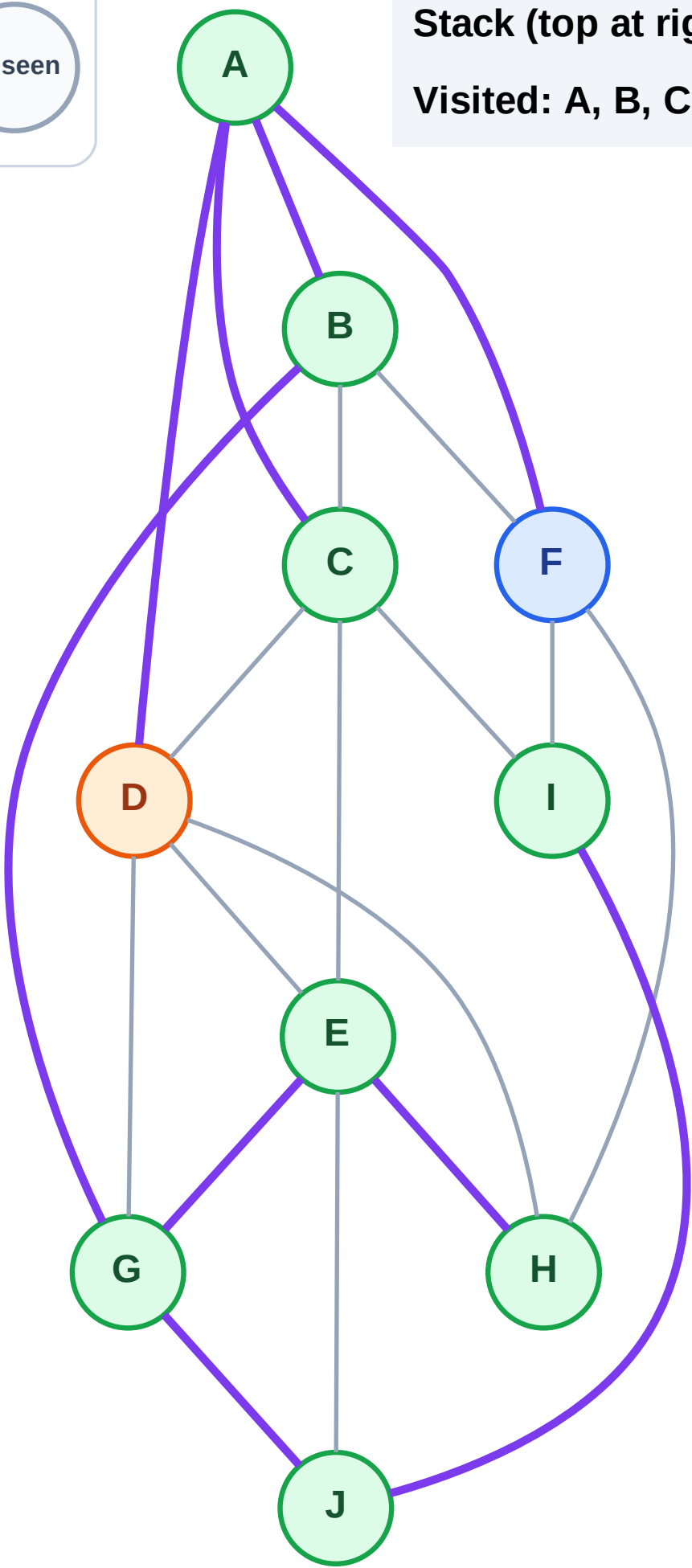
Step 18: Process node D

Legend



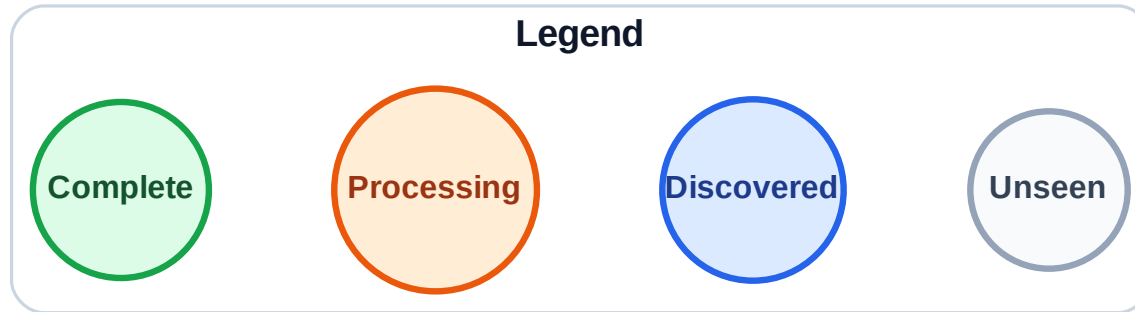
Stack (top at right): F

Visited: A, B, C, E, G, H, I, J



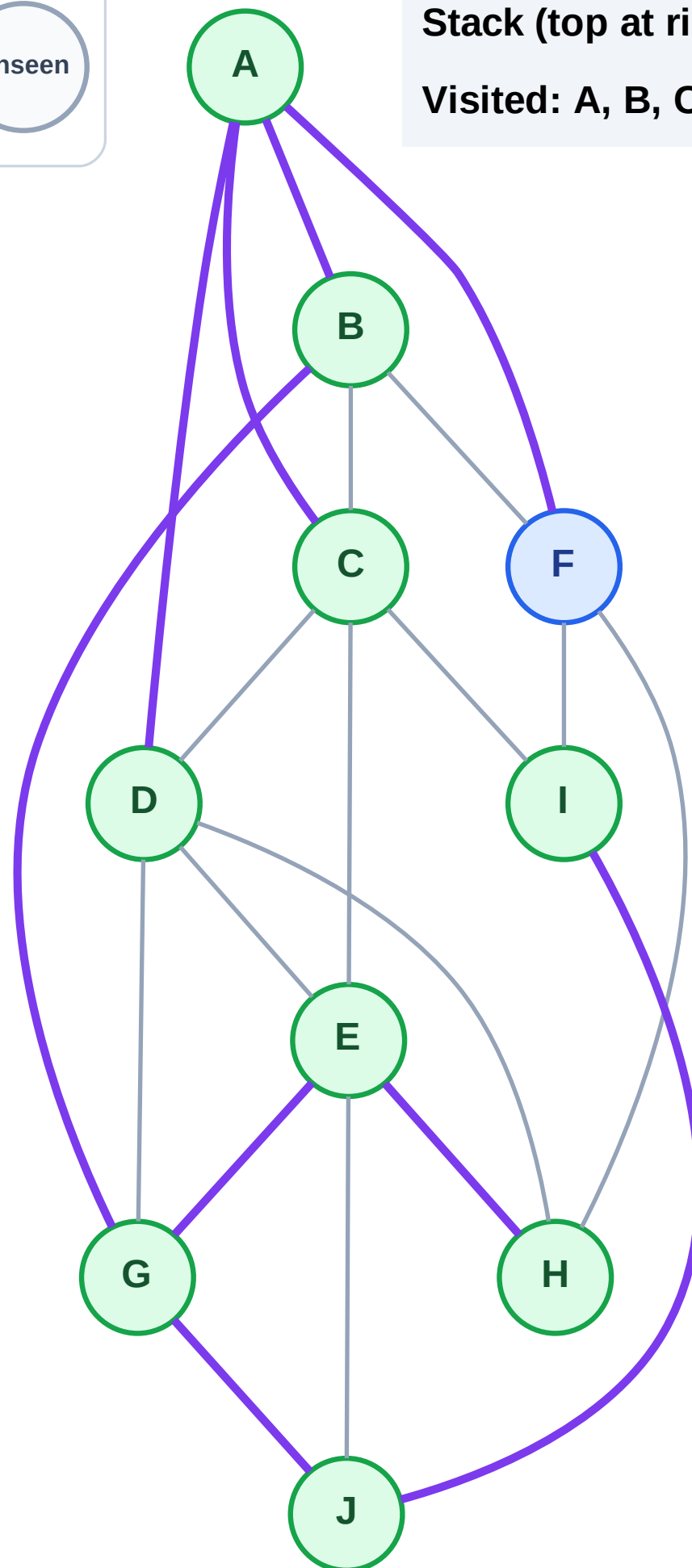
Step 19: No new neighbors from D

Step 19: No new neighbors from D



Stack (top at right): F

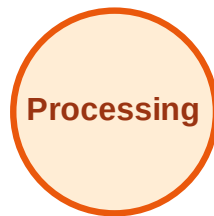
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

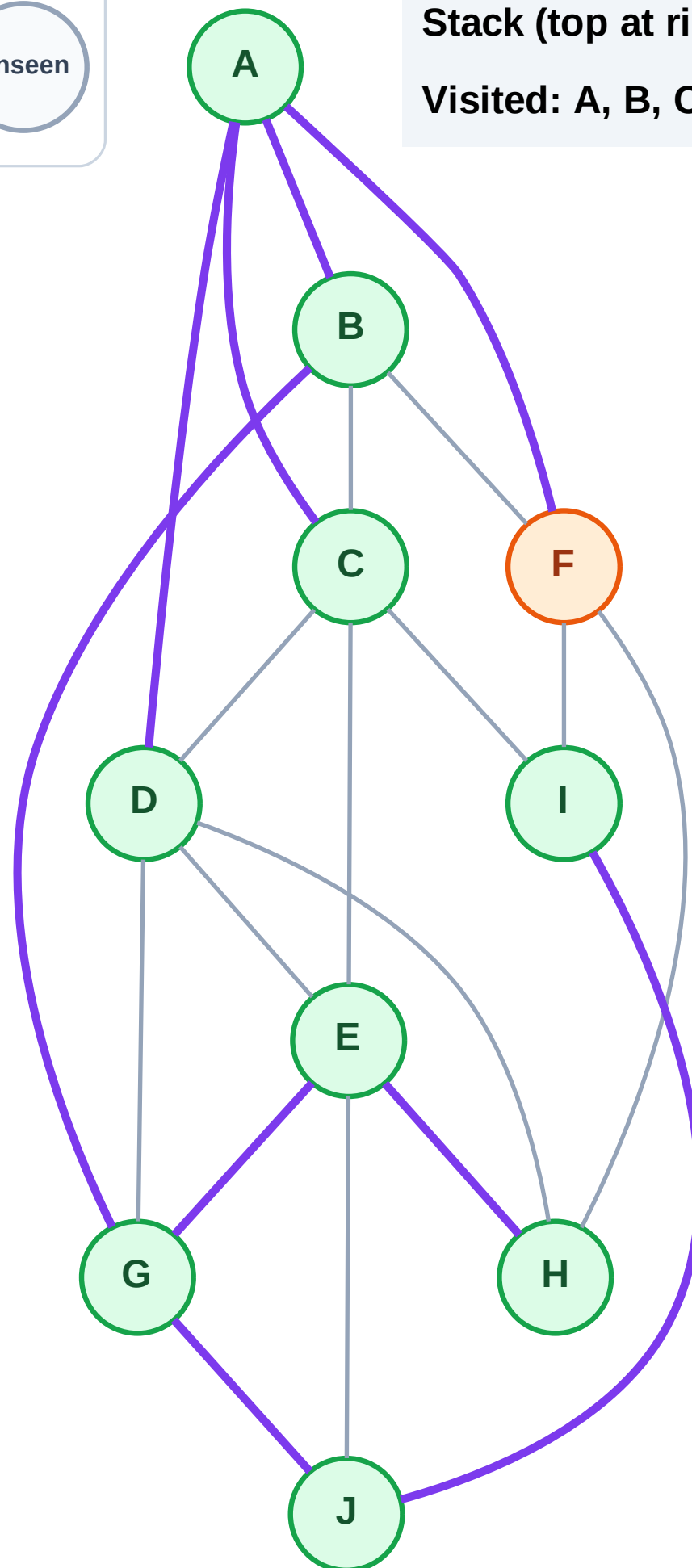
Step 20: Process node F

Legend



Stack (top at right): (empty)

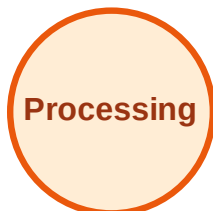
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

